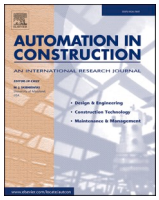
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AI–Digital Twin framework for automated predictive maintenance of bonded insulated rail joints

G. Bianchi *, F. Freddi, F. Giuliani

Department of Engineering and Architecture, University of Parma, 43121 Parma, Italy

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ABSTRACT

Predictive maintenance is a key enabler of automated Structural Health Monitoring (SHM) for safety-critical infrastructure, supporting reliable operation under increasing service demands. This paper presents an integrated Digital Twin–Artificial Intelligence (DT–AI) framework for the automated condition assessment of bonded Insulated Rail Joints (IRJs) and adjacent sleepers. A high-fidelity Digital Twin is developed to simulate joint behaviour under varying preload and support degradation scenarios, generating physically consistent datasets across representative loading conditions. Deep Convolutional Neural Networks (DCNNs) and traditional Machine Learning (ML) classifiers are trained on time–frequency representations of selected predictive indicators to identify and classify degradation states. Results demonstrate that the proposed physics-informed DT-driven framework achieves high diagnostic accuracy and strong generalization capability, with DCNNs models outperforming conventional classifiers. By integrating physics-based simulation with automated data analytics, the proposed framework advances scalable, data-driven SHM and predictive maintenance strategies, for the intelligent automation of railway infrastructure management.

1. Introduction

Modern railway networks are advanced technological systems that underpin sustainable mobility, fostering social cohesion, economic growth, and regional integration [1]. By enhancing safety and capacity while promoting a modal shift from road to rail, efficient rail systems contribute to reduced congestion and greenhouse gas emissions, supporting circular economy objectives [2]. Achieving these goals requires not only infrastructure expansion but also improved reliability, resilience, and optimized lifecycle management of critical assets.

In this context, the digitalization of railway infrastructure, through Digital Twins (DTs), the Internet of Things (IoT), and Artificial Intelligence (AI)-based predictive tools, represents a transformative paradigm for asset management [3]. These technologies enable real-time monitoring, automated anomaly detection, and Predictive Maintenance (PdM), moving beyond traditional time-based scheduling. Their strategic importance is reflected in large-scale national and European initiatives, such as the Italian project MOST — Centro Nazionale per la Mobilità Sostenibile [4], Spoke 4 Railway Transport, funded under the National Recovery and Resilience Plan (NRRP) and the Next Generation EU (NGEU) programme [5], which promote advanced digital solutions for sustainable railway transport.

Among safety-critical railway components, bonded Insulated Rail Joints (IRJs) remain essential in conventional signalling systems until the transition to European Rail Traffic Management System/European Train Control System (ERTMS/ETCS) and Radio Block Center (RBC) technologies, is fully completed. While ensuring electrical insulation and mechanical continuity, IRJs introduce structural discontinuities relative to continuous welded rail, making them inherently susceptible to degradation. Failures may compromise signalling integrity, reduce track stiffness, and adversely affect operational safety. Periodic inspection-based maintenance is increasingly inadequate, as IRJ deterioration results from complex interactions among cyclic loading, bolt preload loss, environmental effects, and support degradation [6].

Despite robust design, bonded IRJs are prone to progressive bolt loosening, cracking, and wear. Loss of bolt preload increases vertical deflection and may induce bolt–hole clearance undetectable until cracking occurs [7]. Their reduced bending stiffness amplifies deformation under train passage, accelerating deterioration and increasing maintenance costs [8,9]. Adjacent sleepers also play a critical role: stiffness discontinuities around joints intensify impact loads and stress concentrations [10], contributing to ballast degradation [11] and

* Corresponding author.
E-mail addresses: giovanni.bianchi@unipr.it (G. Bianchi), francesco.freddi@unipr.it (F. Freddi), felice.giuliani@unipr.it (F. Giuliani).

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potential cracking of prestressed concrete sleepers [12,13]. Although previous studies have documented these degradation mechanisms [14–16], early-stage detection remains challenging with conventional approaches.

Physics-based Digital Twins integrated with data-driven models offer a promising alternative for predictive maintenance [17,18]. While DTs have been applied to railway bridges, switches, and rolling stock [19,20], and previous studies have examined bolt degradation, IRJ behaviour, or sleeper stiffness separately, no work has systematically integrated their coupled interaction within a unified DT–AI framework for automated diagnostics. Traditional methods, such as isolated computer vision or conventional sensors, often rely on heuristic thresholds and manual inspections, limiting scalability and real-time applicability.

To address these limitations, the present study develops a high-fidelity, physics-informed Digital Twin that reproduces the coupled mechanical behaviour of IRJs and adjacent sleepers under both nominal and controlled degradation scenarios that cannot be safely reproduced in the field. DT-generated signals, specifically vertical displacement and gap value. i.e. the longitudinal displacement between adjacent rails, are transformed into time–frequency representations and analysed using Deep Convolutional Neural Networks (DCNNs) and traditional Machine Learning (ML) classifiers for automated condition assessment.

The main contributions of this work are:

- *A unified physics-informed Digital Twin* reproducing the coupled mechanical behaviour of bonded IRJs and adjacent sleepers under varying bolt-preload and support-stiffness conditions.
- *A structured definition of structural health classes* grounded in mechanically interpretable degradation mechanisms.
- *A systematic comparative assessment* of time–frequency-based deep learning models (STFT-CNN and WT-CNN) and a traditional ML decision-tree classifier across six predictive maintenance scenarios.
- *A methodology for quantifying the detectability limits* of various degradation modes, linking classification performance to underlying mechanical phenomena and providing a foundation for future maintenance prioritization.

From an automation perspective, the proposed framework establishes a closed-loop, decision-oriented monitoring workflow in which DT simulations generate labelled degradation scenarios that are mapped to measurable indicators through AI-based classifiers. By embedding mechanical knowledge within the diagnostic logic, the methodology reduces reliance on manual inspections and heuristic thresholds, enabling scalable predictive maintenance aligned with digital transformation objectives. The decision-oriented monitoring workflow reduces human dependency by automating the classification of degradation states, thus enabling data-driven maintenance decisions and mitigating operational bottlenecks throughout the infrastructure lifecycle.

The remainder of the paper is structured as follows: Section 1 provides an overview of the relevant background literature, defines the research gap, and positions the present study. Section 2 describes the DT and AI modelling framework, shown in Fig. 1. Section 3 presents the case studies and comparative analyses, and Section 4 discusses the results in the context of automated predictive maintenance and infrastructure digitalization.

1.1. Digital twins and AI-driven predictive maintenance in railway infrastructure

Digital Twin technology is rapidly advancing in the Architecture, Engineering and Construction (AEC) sector [21], enabling high-fidelity virtual representations of infrastructure under realistic operational conditions. In railway systems, DTs support lifecycle optimization, scenario analysis, and risk assessment [17–20]. However, their full PdM potential emerges when integrated with ML and DL techniques capable

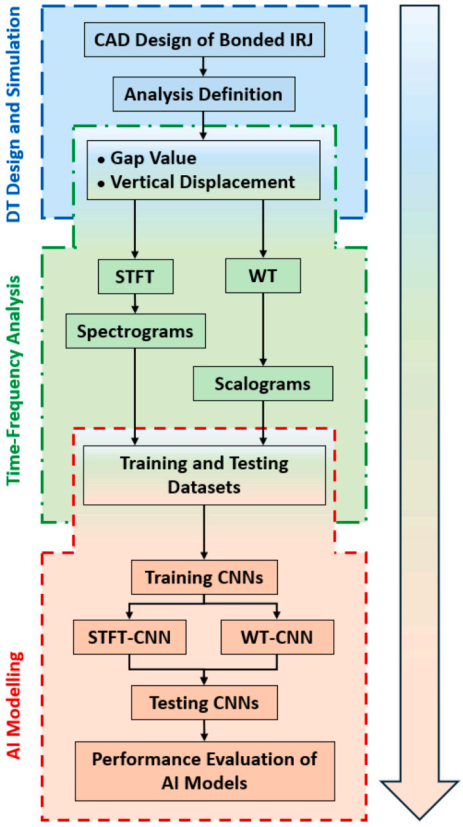


Fig. 1. Framework of the predictive SHM strategy, divided into 3 main interconnected stages.

of extracting patterns from large and heterogeneous datasets [22]. This study introduces a novel integration of a physics-informed Digital Twin with deep learning-based diagnostics, which offers a significant advancement over prior work by addressing the coupled interaction of IRJs and adjacent sleepers.

A key advantage of physics-informed DTs is their ability to generate controlled degradation scenarios that cannot be safely reproduced in the field. Railway datasets are inherently unbalanced: healthy-condition measurements are abundant, whereas severe or progressive damage data are scarce. DT simulations enable systematic generation of labelled, mechanically consistent degradation states, producing training datasets with meaningful predictive content more efficiently than extensive experimental campaigns. When combined with AI models, this capability supports automated scalable SHM workflows while complementing conventional sensor- and vision-based approaches.

AI techniques have demonstrated strong predictive performance across multiple domains [23,24] and are increasingly adopted in railway monitoring [25,26]. While interpretable ML methods provide structured modelling of multimodal sensor data, CNN-based DL approaches enable automatic feature extraction from high-dimensional signals, improving scalability and reducing manual feature engineering. To ensure interpretability and operational reliability, we embed mechanical knowledge into the DCNN framework, guiding the network with physically meaningful features.

Large-scale implementations further demonstrate the potential of AI-enhanced DTs in rail systems. For example, Siemens-supported projects in China's high-speed rail network employed ML-augmented DT frameworks to improve reliability and maintenance efficiency [27]. Other studies explored Ground Penetrating Radar (GPR) and Interferometric Synthetic Aperture Radar (InSAR) data integration for predictive maintenance [28], HybridGAN architectures for defect detection under low-quality imagery [29], and UAV-based monitoring for safety

enhancement [30]. Despite these advances, many studies treat simulation, sensing, and AI diagnostics as partially disconnected elements rather than components of a fully integrated, physics-informed workflow.

1.2. Degradation mechanisms and monitoring of bonded IRJs and sleepers

Historically, maintenance of bonded IRJs has relied on periodic inspections and preventive routines [14]. However, IRJs remain vulnerable to bolt loosening, cracking, and material degradation. Reduced bending stiffness amplifies dynamic response under loading [8,9], while preload loss increases vertical displacement and gap variation preceding visible damage [7]. Stiffness discontinuities at joints intensify impact forces on adjacent sleepers [10], contributing to ballast degradation [11], increased rail-seat pressures, and potential cracking of prestressed concrete sleepers [12,13]. The coupled interaction between joint flexibility and sleeper support conditions critically affects track durability and geometry preservation.

Modern monitoring methods, including axlebox acceleration-based drive-by monitoring [31], Fibre Bragg Grating (FBG)-based curvature sensing [11], acoustic emission techniques [32], and non-destructive evaluation approaches such as ultrasonics, infrared thermography, and ML-enhanced vision systems [12,33], provide valuable diagnostic information. However, these techniques require extensive field instrumentation and do not allow controlled reproduction of severe degradation states for algorithm training.

1.3. Time–frequency representations and CNN-based diagnostics

Transforming time-series signals into time–frequency representations, such as spectrograms (via Short-Time Fourier Transform, STFT) and scalograms (via Wavelet Transform, WT), enables mapping structural responses into image-like formats suitable for CNN analysis.

Spectrogram-based CNNs have outperformed traditional feature-engineering methods in bearing diagnostics [34], pavement crack detection [35], and railway axle fault identification [36]. These approaches demonstrate that time–frequency imaging allows CNNs to capture subtle spectral signatures associated with incipient damage.

For bonded IRJs, indicators such as vertical displacement and gap value, i.e. the longitudinal displacement between adjacent rails, previously identified as predictive of structural degradation [16,37], exhibit transient spectral characteristics during wheel passage. Converting these signals into high-resolution time–frequency maps enables automated extraction of degradation patterns that may remain undetectable in raw time-domain analysis.

1.4. Research gap and positioning of the present study

DT modelling, FE simulation, and AI-based diagnostics are established in SHM [35,38]. In railway engineering, DT and AI approaches are increasingly applied to individual subsystems [19,20]. Recent work has proposed DT-based frameworks for enhancing the resilience of cyber-physical systems [22]. Systematic reviews have synthesised ML methods for predictive maintenance applications [23], outlined the foundations of AI for engineering practice [24], and highlighted emerging ML-driven approaches for SHM under high-dimensional data environments [25]. More recently, railway-specific reviews have documented the transition from conventional monitoring techniques toward predictive maintenance strategies [26]. However, the coupled interaction between bonded IRJs and adjacent sleepers has not been investigated within a unified DT–AI framework designed for automated condition assessment.

Building on our previous work [16], which combined a DT with a decision-tree classifier under simplified assumptions, this study advances modelling fidelity and automation by capturing IRJ–sleeper coupling and exploiting full time-series signals transformed into time–frequency representations for DCNN-based diagnostics.

The proposed framework is designed as a physics-informed DT, capable of generating mechanically consistent and systematically labelled degradation states to address the scarcity of real failure data. While the current validation is conducted on synthetic datasets, the architecture is conceived for future integration with laboratory and field measurements through transfer learning and hybrid virtual–physical updating strategies. Unlike traditional FE models, this DT-oriented framework is intended to integrate real-time data streams and provide dynamic feedback for decision-making, bridging the gap between numerical simulation and operational predictive maintenance.

The primary objective is the development and validation of a coherent DT–AI synergy framework for automated predictive maintenance. The comparative evaluation of deep learning and traditional ML classifiers serves to assess the robustness and scalability of this integrated architecture rather than to perform isolated algorithm benchmarking. The novelty lies in the systematic integration of validated modelling and AI tools into a physics-informed, decision-oriented workflow that supports scalable automation and future alignment between virtual simulations and real-world railway operation.

2. Materials and methods

This section presents the integrated methodology underpinning the proposed DT–AI framework, combining a physics-informed Digital Twin for modelling the coupled behaviour of bonded IRJs and adjacent sleepers (Section 2.1) with DCNN-based time–frequency analysis for automated predictive maintenance and condition classification (Section 2.2).

2.1. Digital Twin framework for bonded IRJ and sleeper modelling

The proposed Digital Twin (DT) reproduces the coupled mechanical behaviour of bonded Insulated Rail Joints (IRJs) and adjacent sleepers under controlled degradation scenarios, including bolt-preload loss and sleeper stiffness reduction. It generates physically consistent synthetic data suitable for AI-driven diagnostics, supporting predictive maintenance without requiring exhaustive field measurements in the early development stages. Previous work demonstrated that classifiers trained on synthetic DT-generated signals generalize effectively to safe-field experimental measurements, providing evidence that the Digital Twin reproduces mechanically relevant responses that reliably enable AI-driven diagnostics [16].

2.1.1. Geometrical configuration and modelling assumptions

The DT of the suspended bonded IRJ was developed using the FE platform Abaqus/CAE. The model provides a geometrically simplified

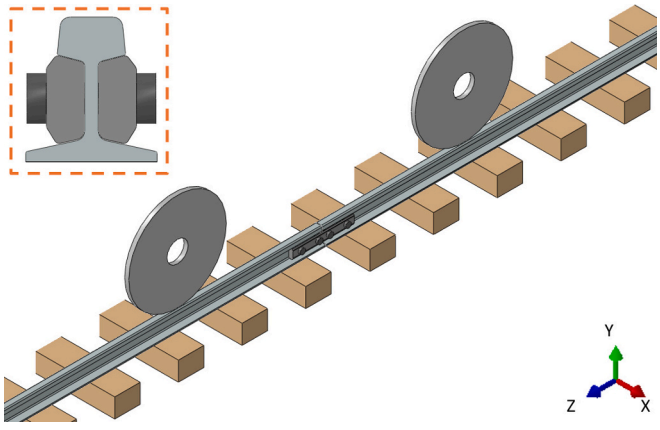


Fig. 2. Cross section of IRJ in orange box; digital twin of bonded IRJ and two wheels.

yet mechanically representative configuration of a 50 UNI rail profile (Fig. 2), including joint components compliant with Rete Ferroviaria Italiana (RFI) standards. Two 9 m rail segments are connected through fishplates and four bolts, forming an 18 m assembly sufficient to reproduce the global structural behaviour of a finite rail modelled as a beam on an elastic foundation.

Electrical insulation between the rails is represented by a 5 mm end post located within the rail gap. Secondary insulating elements (bushings and fishplate interlayers) are omitted, as supplementary analyses confirmed that their inclusion introduces significant numerical nonlinearity while exerting negligible influence on global mechanical response. The adopted configuration therefore preserves structural fidelity while ensuring numerical robustness.

The passage of two train wheels (diameter 920 mm, wheelbase 2700 mm) is simulated following established IRJ methodologies [39]. Wheel loads are applied vertically and symmetrically along the rail axis, consistent with straight-track conditions. Minor-axis bending and torsional effects are neglected, as previous IRJ studies identified vertical load-transfer mechanisms as the dominant contributors to gap formation and rail-end displacement [40,41].

2.1.2. Material modelling, contact formulation, and loading strategy

All structural components, rails, fishplates, bolts, wheels, and end post, are explicitly modelled and assembled (Fig. 2). Material properties and cross-sectional parameters are derived from literature values for standard rail steels and commonly adopted insulating materials (nylon, polyurethane, epoxy–fibreglass composites) [42,43].

The analysis adopts a quasi-static formulation with geometric nonlinearity enabled (Nlgeom option). Bolt preload and wheel loads are applied incrementally to ensure numerical stability. Wheel passage is simulated through prescribed translational motion at a temporal resolution equivalent to 1000 Hz.

Rail–end post interfaces are defined via tie constraints [44], while all remaining interactions are governed by a general contact formulation with a tangential penalty coefficient of 0.6, representative of dry wheel–rail contact conditions [16,45]. This configuration captures the dominant nonlinear contact phenomena influencing joint stiffness and gap evolution.

A constant wheel speed of 18 m/s (≈65 km/h) is adopted, representative of regional and suburban operating regimes in which bonded IRJs remain widely deployed [37,46]. Within this moderate-speed range, quasi-static modelling reproduces the primary stiffness-governed response with limited deviation from fully dynamic peak indicators [47], while maintaining computational efficiency compatible with large parametric campaigns. Extension to variable-speed and dynamic analyses is identified as a natural progression of the framework.

2.1.3. Elastic foundation modelling and degradation parameterisation

The rail is modelled as a beam supported by an elastic foundation representing the combined stiffness of sleepers and ballast. Standard monoblock sleepers are discretised into 240 mm × 135 mm support segments at 600 mm spacing, resulting in 15 sleepers per rail (30 total) [48]. The elastic foundation is calibrated from a representative track stiffness of 21 N/mm^2 for conventional ballasted systems [40,41], consistent with benchmark parameters adopted in previous track-modelling studies [49]. This value corresponds to a total nominal foundation stiffness per area of 0.39 N/mm^3 distributed along the 18 m rail length. Consistent with the adopted discretization into 30 sleeper supports, this global stiffness is uniformly partitioned, resulting in an equivalent per-sleeper foundation stiffness per area of 0.013 N/mm^3 under intact conditions. Sleeper degradation is modelled as a reduction in foundation stiffness per area. Sleeper degradation is modelled as a 50% reduction of this local support stiffness (0.0065 N/mm^3), representing significant but realistic deterioration associated with ballast fouling, void development [11], or moisture-induced softening documented in railway engineering literature [12,13]. This value does not represent a universal maintenance threshold; rather, it defines a mechanically interpretable degradation magnitude enabling systematic evaluation of detectability limits within the DT–AI framework.

Sleepers are labelled sequentially (S1–S30, Fig. 3), with particular focus on those adjacent to the joint (S15–S16) and those located near mid-span regions (S8 and S23). This spatial configuration enables investigation of degradation proximity effects on structural response.

2.1.4. Bolt preload modelling and structural health states

Bolt assemblies are represented through a simplified but mechanically consistent modelling strategy in which the bolt–nut–washer system is treated as a unified preloaded component [40,43]. This approach captures the dominant influence of clamping force on joint flexural stiffness and contact pressure redistribution without introducing unnecessary local nonlinearities.

A nominal preload of 75 kN represents standard assembly conditions. Reduced preloads of 50 kN and 25 kN simulate progressive clamping-force relaxation due to cyclic loading, embedment, and micro-slip phenomena observed in service conditions [7]. Missing-bolt configurations represent extreme loss of mechanical continuity.

These preload levels are not arbitrarily selected numerical partitions; they correspond to mechanically distinct regimes characterised by monotonic amplification of gap values and rail-end vertical displacement. Structural health states are therefore defined according to transitions in load-transfer behaviour rather than abstract signal thresholds.

2.1.5. Sensitivity-informed parameter selection

The degradation parameters adopted in the Digital Twin are defined following a sensitivity-informed strategy aimed at ensuring mechanical representativeness and robustness of the diagnostic framework. Parameter ranges are selected to span distinct regimes of load-transfer behaviour, enabling clear discrimination of degradation-induced structural responses rather than reproducing a single calibrated condition.

The selected bolt preload levels (75–25 kN) reflect progressive clamping-force loss mechanisms reported in bonded IRJs [16]. Similarly, the 50% reduction in sleeper support stiffness is consistent with degradation scenarios associated with ballast fouling and void development [11] in railway track systems [12,13]. These values are therefore interpreted as physically meaningful degradation states rather than prescriptive thresholds.

Targeted sensitivity checks indicate that the primary diagnostic indicators, namely gap value and vertical displacement, vary monotonically with stiffness degradation, while secondary modelling parameters exert limited influence on global response patterns. This behaviour is consistent with established vehicle–track interaction studies, which identify support stiffness as the dominant factor governing track response [50]. Consequently, class separability is driven by intrinsic mechanical contrasts rather than specific parameter tuning.

This parameterization provides a robust basis for DT-generated training datasets, supporting future integration with experimental data within hybrid virtual–physical monitoring frameworks.

2.1.6. Numerical verification and experimental benchmarking

A mesh convergence study was conducted by refining discretisation in critical regions (rail ends, bolt interfaces, fishplate contacts) and monitoring peak gap and vertical displacement under a 60 kN load. Variations below 2% beyond the adopted C3D8R mesh confirmed discretisation-independent global response. Reduced-integration 8-node brick elements provide an effective balance between accuracy and computational cost.

The modelling framework builds upon established finite-element formulations for bonded and mechanical rail joints, including analyses of IRJ deflection behaviour under wheel loading [39,43], bolt-preload effects and stress concentrations in rail joints [40,41], and train–track interaction mechanisms associated with insulated joints [48]. Experimental validation of IRJ response through dedicated field measurement

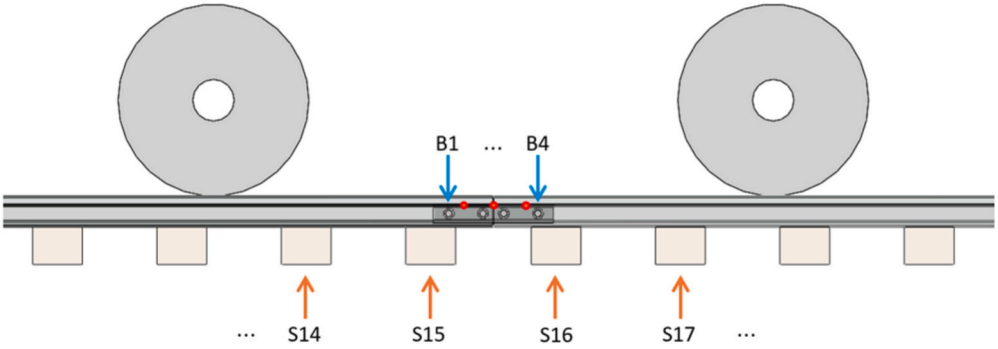


Fig. 3. Frontal view of the bonded IRJ, with bolts numbered B1–B4 and red probes marked on the surface. Sleepers near the IRJ, numbered S14–S17, are highlighted, with the full set numbered sequentially from S1 to S30. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

campaigns has also been reported in the literature [15], providing a benchmark for assessing the mechanical fidelity of the adopted modelling approach. In those studies, FE-predicted gap values during wheel passage showed strong agreement with field measurements, with deviations consistently remaining below critical tolerance levels across all investigated load conditions. These results support the mechanical fidelity of the adopted load-transfer representation.

Furthermore, prior work demonstrated that a classifier trained exclusively on DT-generated synthetic gap signals successfully generalised to safe-field experimental measurements [16], providing preliminary evidence of transferability from simulation to practice.

2.1.7. *Synthetic datasets as a structured baseline for automation*

The DT configuration remains consistent across all case studies presented in Section 3; only degradation parameters (bolt preload and sleeper stiffness) are varied. Simulation outputs, specifically gap value and rail-end vertical displacement extracted at probe locations near the joint and critical sleepers (Fig. 3), are compiled into labelled datasets for AI training and testing.

The reliance on synthetic data is a deliberate methodological choice. Severe degradation scenarios are inherently scarce, unsafe to reproduce experimentally, and typically imbalanced relative to healthy conditions. The Digital Twin provides a controlled, mechanically consistent environment in which primary degradation mechanisms can be systematically isolated and parameterised.

The present validation therefore establishes a physics-informed upper-bound benchmark under controlled conditions. Environmental variability, measurement noise, and operational uncertainty are not explicitly modelled at this stage but are identified as part of ongoing hybrid virtual–physical updating strategies aimed at progressively enhancing robustness under real-world deployment conditions.

In this context, the use of synthetic data is not intended to approximate full operational complexity, but to establish a controlled, mechanistically interpretable learning baseline from which model behaviour can be rigorously assessed before progressive exposure to real-world variability.

2.2. *DCNN with time–frequency analysis*

Recent advances in AI-driven railway monitoring have demonstrated the value of combining high-fidelity datasets with data-driven models for damage detection and prognosis [25,26]. Similar trends have also emerged in broader SHM applications [51,52]. Within this context, DTs are particularly effective for SHM, as they provide physically consistent data under both nominal and systematically degraded conditions. Since no single AI technique is universally optimal, this study adopts and compares multiple learning strategies to evaluate their suitability for automated health assessment of bonded IRJs.

Building on previous Machine Learning (ML) investigations on bolt preload degradation in IRJs [16], the present work extends the framework to Deep Learning (DL) approaches inspired by recent fault-classification methodologies based on time–frequency imaging [34–36]. The proposed methodology consists of three sequential stages: (i) transformation of DT-generated time-domain signals into time–frequency–energy representations, (ii) training of Convolutional Neural Networks (CNNs) on the resulting image-based datasets, and (iii) performance testing for automated classification of IRJ structural health states.

2.2.1. *Training data and time–frequency feature construction*

Vertical displacement and gap value, defined as the relative longitudinal displacement between adjacent rails, were selected as primary diagnostic variables. These indicators exhibit strong mechanical sensitivity to bolt preload loss and sleeper stiffness degradation, enabling physically interpretable definition of structural health classes. Their relevance is supported by previous numerical and experimental investigations [15,16] and further demonstrated in Section 3. Moreover, both quantities are measurable in practice using modern sensing technologies [53], reinforcing the translational potential of the proposed framework.

For each of the six predictive maintenance case studies, DT outputs were organised into labelled datasets representing discrete degradation states. Each sample includes the corresponding wheel load, which is explicitly incorporated into the neural architecture. This multimodal configuration enables the classifier to distinguish load-induced variations from degradation-induced effects, reducing the risk of misclassifying high-load healthy states or low-load damaged states. Explicit inclusion of operational load information avoids the need for substantially larger datasets to implicitly learn load effects from image features alone.

Each simulation covers 0.8 s, corresponding to wheel traversal of the joint at 18 m/s. Signals were recorded at 1000 Hz, resulting in 800 time samples per simulation. Time-domain responses were transformed into time–frequency representations using both the Short-Time Fourier Transform (STFT) and the Wavelet Transform (WT), generating spectrograms and scalograms, respectively. Spectrograms provide fixed resolution suitable for quasi-stationary components, while scalograms exploit multi-resolution analysis to capture localized non-stationary phenomena. These representations allow CNNs to automatically extract degradation signatures without manual feature engineering.

The STFT implementation employed a Hamming window of approximately one quarter of the signal length with 50% overlap. The FFT length was defined as the maximum between 256 and the next power of two of the window size, ensuring adequate frequency resolution while limiting spectral leakage. This configuration provided a balanced compromise between temporal localisation of wheel–joint

interaction effects and frequency discrimination of stiffness-related spectral shifts.

For the WT-based scalograms, a filter bank was constructed for the full 800-sample signal length. A discretisation of 12 voices per octave was adopted to enhance frequency resolution while maintaining computational efficiency. The analytic Morse wavelet was selected for its robustness in capturing localized amplitude and frequency variations in non-stationary mechanical signals. Boundary artefacts were mitigated via signal reflection at the edges.

Preliminary sensitivity checks were conducted on a reduced subset of simulations by varying window length, overlap ratio, and analytic wavelet selection within the same family. These variations did not qualitatively alter class separability or relative model ranking, confirming that diagnostic performance is not an artefact of singular parameter tuning. Since the dataset is synthetically generated in a controlled numerical environment without measurement noise, time–frequency representations reflect exclusively the mechanically modelled degradation mechanisms. The objective of this study is therefore not optimisation of imaging parameters, but validation of a physics-informed DT–AI integration framework. Extended sensitivity analyses will become particularly relevant when synchronizing the DT with laboratory and field measurements incorporating operational variability.

An 80/20 stratified split between training and testing datasets was adopted for each case study, ensuring balanced representation of all health classes. The total number of samples per scenario is reported in Table 1 (Section 3). The 20% hold-out subset was reserved exclusively for evaluation on previously unseen data.

The STFT and WT parameters were selected based on established time–frequency analysis methodologies for non-stationary structural signals [34–36], balancing temporal and frequency resolution to capture localized degradation effects while maintaining computational efficiency [54,55]. Preliminary sensitivity analyses confirmed that moderate variations in window length, overlap, and wavelet choice did not materially affect class separability, consistent with prior studies on vibration-based damage detection in railway components [56].

2.2.2. AI classification models

All models were developed within the MATLAB® environment, which integrates signal processing, image analysis, and machine learning toolboxes within a unified SHM-oriented workflow [57,58]. DT simulation outputs were transformed into image-based representations and processed using Deep CNN (DCNN) architectures designed to accommodate both image inputs and operational features.

The adopted network consists of an image input layer matching spectrogram or scalogram dimensions, followed by a convolutional layer with 5×5 kernels and 32 filters. Batch normalisation and a rectified linear unit (ReLU) activation are applied to enhance convergence stability. A fully connected layer with 50 neurons produces a compact embedding of extracted features. The resulting feature vector, originally in "SSCB" (spatial, spatial, channel, batch) format, is flattened, into "CB" (channel, batch) format, and concatenated with the numerical wheel-load input along the channel dimension. Final classification is performed through a fully connected layer with output size equal to the number of health states, followed by a softmax layer to generate class probabilities. A schematic representation is provided in Fig. 4.

Training was conducted using Stochastic Gradient Descent with Momentum (SGDM) and cross-entropy loss. The training configuration includes an initial learning rate of 1×10^{-4}, mini-batch size of 15, and a maximum of 20 epochs. Convergence stability was monitored during training, and overfitting was mitigated through conservative hyperparameter selection and balanced dataset partitioning. No adaptive learning-rate scheduling was applied, as stable convergence was consistently achieved under these settings.

The architectural depth was intentionally limited. The objective of this work is not benchmarking increasingly complex deep networks, but validating a deployable DT–AI framework suitable for predictive maintenance automation. Excessively deep architectures may increase computational burden and reduce interpretability without proportionate diagnostic gains under mechanically structured datasets. The selected configuration provides sufficient representational capacity to capture localized degradation patterns while remaining computationally efficient and compatible with edge- or cloud-based monitoring infrastructures.

Two DL models were implemented using this architecture: STFT-CNN and WT-CNN. In parallel, a supervised ML model was developed using time-domain DT outputs. A medium-depth Decision Tree classifier was selected as a transparent and computationally efficient baseline. Because ML approaches require explicit feature engineering, gap value and vertical displacement were directly used as input features. This structured comparison enables assessment of feature-learning DL versus feature-engineered ML within the same physics-informed dataset, supporting evaluation of scalability, interpretability, and automation readiness rather than exhaustive classifier benchmarking.

To further corroborate the robustness of the proposed framework, an additional reference model based on a Support Vector Machine with radial basis function (RBF) kernel was also implemented. The corresponding results are reported in Section 2 of Supplementary Material.

2.2.3. Model validation strategy and generalization control

Dataset partitioning followed an 80/20 stratified hold-out approach for each predictive maintenance scenario. The testing subset was strictly excluded from training and used solely for final generalization assessment on previously unseen data. Given the mechanically consistent

Table 1
Main characteristics of the six PdM case studies.

Predictive Model	Objective	Predictive Classes	Predictive Parameters	Wheel-loads levels (kN)	Dataset size (samples)
3.1	Model for predicting bolt preload conditions in a bonded IRJ under 3 wheel-loading scenarios	4 Classes based on bolt preload conditions	Gap Value Vertical Displacement	60/90/120	360
3.2	Model for identifying bolt configurations in the "warning" class of loosening	5 Classes based on loosed bolt configurations	Gap Value Vertical Displacement	60	150
3.3	Model for predicting SH conditions based on bolt loosening and sleeper stiffness loss adjacent to the bonded IRJ	6 Classes based on bolt preload and/or sleeper stiffness lost	Gap Value Vertical Displacement	60/90	600
3.4	Model for predicting the loss of sleeper stiffness in the vicinity of the bonded IRJ	6 Classes based on sleeper stiffness lost	Gap Value Vertical Displacement	60/90	480
3.5	Model for predicting the loss of stiffness in Sleeper 8 by measuring vertical displacement at 3 different positions	2 Classes based on the loss of stiffness in Sleeper 8	Vertical Displacement measured at sleepers 8, 11, and 14	60/90	300
3.6	Model for predicting the loss of stiffness in Sleeper 8 using the vertical displacement profile of the entire railway section at 5 different time instants	2 Classes based on the loss of stiffness in Sleeper 8	Railway vertical displacement Profile at 5 significant time instants	60/90	300

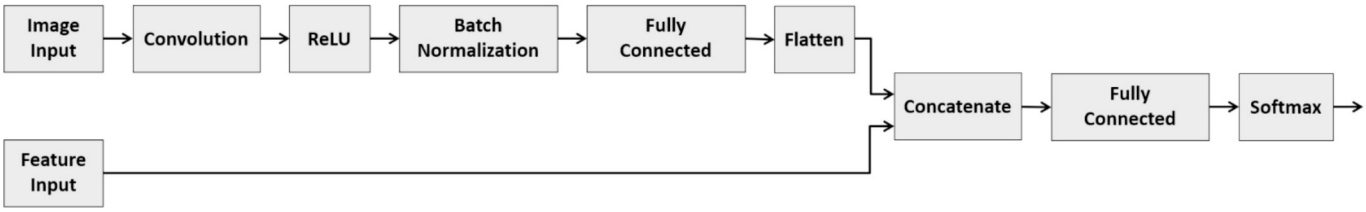


Fig. 4. Schematic framework of the neural network.

nature of the DT simulations, the datasets exhibit high predictive coherence and well-defined class separability, enabling stable convergence with moderate dataset sizes.

Overfitting was mitigated through complementary strategies. For the DCNN, a lightweight architecture was adopted (32 convolutional filters, fully connected layer of size 50, MaxEpochs = 20, mini-batch size 15) to balance representational capacity and computational efficiency. During training, convergence behaviour was monitored using MATLAB's *ValidationFrequency* option, which evaluates model performance at fixed intervals of 10 iterations (mini-batch updates). Importantly, these evaluations are performed on mini-batches drawn from the training data stream and are used exclusively for training diagnostics (i.e., tracking loss and accuracy evolution). They do not constitute an independent validation dataset and are not used for hyperparameter tuning or model selection.

Under this configuration, generalization is assessed through two complementary mechanisms: (i) iterative monitoring of training dynamics (loss and accuracy trends across mini-batches), ensuring stable convergence without divergence or memorisation effects; and (ii) final evaluation on the independent 20% hold-out test set, which provides an unbiased estimate of predictive performance.

For the decision-tree model, 5-fold cross-validation, consistent with the approach adopted in [16], was additionally implemented within MATLAB's Classification Learner app to provide a standard robustness check.

A separate static validation set was not introduced as an additional partition. In deep learning workflows, validation sets are primarily required for hyperparameter optimisation and model selection [55,59]. In the present study, model architecture and training parameters are defined a priori and not iteratively tuned against validation performance. Moreover, the datasets are synthetically generated, physics-informed, and noise-free, resulting in deterministic class structure and reduced variance compared with typical data-driven applications. Under these conditions, a two-way hold-out strategy provides a statistically consistent and computationally efficient framework for assessing generalization without introducing unnecessary data fragmentation.

The adopted 80/20 split remains consistent with established machine learning practice for controlled datasets and provides an effective balance between training sufficiency and unbiased evaluation [60,61].

Because the datasets are synthetically generated and noise-free, the reported performance represents an upper-bound benchmark of intrinsic mechanical separability. Equivalent performance under real operating conditions cannot be assumed and will require progressive integration of noise-contaminated and field-acquired data through hybrid virtual–physical updating strategies.

2.2.4. Performance testing

Model performance was evaluated using standard classification metrics, including accuracy (A), precision (P), recall (R), and F1-score (F1), as defined in [16]. Accuracy quantifies overall correctness, precision assesses reliability of positive predictions by limiting false positives, recall measures detection completeness by minimizing false negatives, and F1-score balances precision and recall.

Performance assessment was further supported by confusion matrices and receiver operating characteristic (ROC) curves. ROC analysis illustrates the trade-off between true positive rate (TPR) and false positive rate (FPR) across decision thresholds, while the area under the curve (AUC) provides a scalar measure of discrimination capability, with values approaching unity indicating superior separability [16]. These evaluation criteria are widely adopted in SHM and related diagnostic domains [35].

The integrated MATLAB® workflow enables consistent preprocessing, training, and evaluation within the same computational environment, ensuring methodological reproducibility and transparent comparison between DL and ML approaches.

3. Results

The six case studies presented below progressively address increasing levels of diagnostic complexity, from global bolt preload assessment to coupled bolt–sleeper degradation and remote support monitoring. The 6 predictive maintenance (PdM) strategies investigated, are summarized in Table 1 and detailed in the following subsections. They differ primarily in their PdM objectives: assessing joint integrity under varying bolt preload conditions and evaluating sleeper-stiffness degradation near the rail joint. Cases 3.1 and 3.2, which address the PdM of bonded IRJs under variable bolt-preload conditions, constitute a natural extension of the SHM framework originally introduced in [16]. Unlike the earlier formulation, where the DT was coupled solely with a ML algorithm, these cases broaden the framework to accommodate more complex operational scenarios.

Following the development of the DT of the bonded IRJ (Section 2.1), FE simulations are performed to generate the datasets used in each predictive approach. The number of simulations per model is reported in Table 1, with additional comments provided where relevant. Each dataset is split into training (80%) and testing (20%) subsets, ensuring balanced class distributions and wheel-load levels to support robust AI-model training. In Cases 3.1–3.4, gap values and local vertical displacements near the joint serve as predictive indicators of structural condition. In contrast, Cases 3.5–3.6 employ alternative monitoring strategies that rely solely on vertical displacement.

The resulting data are processed for labelling and conversion into time-frequency representations, spectrograms and scalograms, used to train and test the STFT-CNN and WT-CNN, respectively, or for preprocessing prior to input to the ML classifier. Table 1 also lists the number of predictive classes and corresponding labels, which are fully described in the individual subsections. Briefly, the safe class represents IRJs with correct bolt preload and nominal sleeper conditions; classes marked B* identify loosened-bolt configurations, while classes marked S* denote sleepers with stiffness degradation (Fig. 3).

Model performance is evaluated using the metrics defined in Section 2.2.3. Across all applications, the predictive models demonstrate high accuracy, confirming both the maturity and practical relevance of the proposed methodologies. These results further emphasize the strategic importance of advancing digitalization and smart maintenance strategies for railway infrastructure. A comprehensive discussion of the results is presented in Section 4.

Additional results obtained using an SVM–RBF reference model are reported in Section 2 of Supplementary Material for completeness.

3.1. Bolt preload assessment under multiple wheel loads

This case study investigates the capability of the proposed DT–AI framework to identify progressive bolt-preload degradation in bonded IRJs under varying operational loading conditions.

3.1.1. Predictive objective and dataset preparation

The model evaluates the structural health of bonded IRJs by analysing bolt preload conditions under three wheel-loads (60, 90, and 120 kN). Digital Twin simulations yielded a balanced dataset of 360 samples based on two predictive indicators: gap values and vertical displacements near the IRJ.

Four structural health classes were defined based on mechanically interpretable degradation states derived from the physics-informed Digital Twin, consistent with the framework introduced in [16]. Class boundaries were established by progressively reducing bolt preload and analysing the resulting changes in joint stiffness, gap value, and vertical displacement under wheel passage. The safe class corresponds to the nominal assembly condition (75 kN preload per bolt). The warning and risk classes represent controlled reductions to approximately 66.6% and 33.3% of nominal preload, respectively, modelling progressive clamping-force loss and the associated reduction in joint flexural stiffness. The fault class represents severe degradation with two missing bolts, producing marked alterations in load-transfer mechanisms.

Multiple bolt layouts were analysed (e.g., central bolts B23, outer bolts B14, alternating pairs B13/B24, individual-rail connections B12/B34, and the full set B1234). As supported by Table 3, Figs. S1–S2, and the findings of [16], these configurations produce similar feature signatures within each class, justifying their grouping. A more granular classifier could detect specific loosened configurations but would require additional training data, as explored in Section 3.2.

Digital Twin simulations revealed monotonic, mechanically consistent amplification of gap values and vertical displacement with decreasing preload, confirming progressive reduction of joint bending stiffness and altered load-transfer mechanisms. These distinct response patterns provide a robust basis for defining discrete structural health classes suitable for supervised learning.

Figs. 5a and 6a show representative gap and vertical displacement responses under a 60 kN wheel load. Fig. 5b–e presents spectrograms and scalograms of gap values for safe and risk conditions, while Fig. 6b–c shows time–frequency analyses of vertical displacement in the safe case. Numerical values at four critical wheel positions, P1 (first wheel on the joint), P2 (wheels symmetric about the joint centreline), P3 (second wheel on the joint), and P4 (residual displacement after both wheels at 0.7 s), are reported in Table 2a–c and were used as ML training inputs [16]. Both gap and vertical displacement increase monotonically with decreasing preload, with the largest deviations occurring at P1 and P3 during direct wheel–joint interaction. At P2, the fault condition exhibits distinct amplification relative to other classes.

It is emphasized that the purpose of this classification is not to establish field-ready maintenance thresholds, which would require extensive experimental calibration and regulatory validation, but rather to provide a structured and physically grounded representation of progressive degradation states for DT–AI framework development.

3.1.2. Evaluating predictive performance

Classifier performance was evaluated using confusion matrices, where diagonal entries indicate correct classifications [16,58]. Figs. 7 and 8 show matrices for STFT-CNN, WT-CNN, and Decision Tree classifiers trained on gap values and vertical displacements, respectively, while Table 3a–b summarize associated performance metrics. Across all models and inputs, predictive accuracy exceeds 90%, confirming both the diagnostic relevance of the indicators and the robustness of the DT–AI framework.

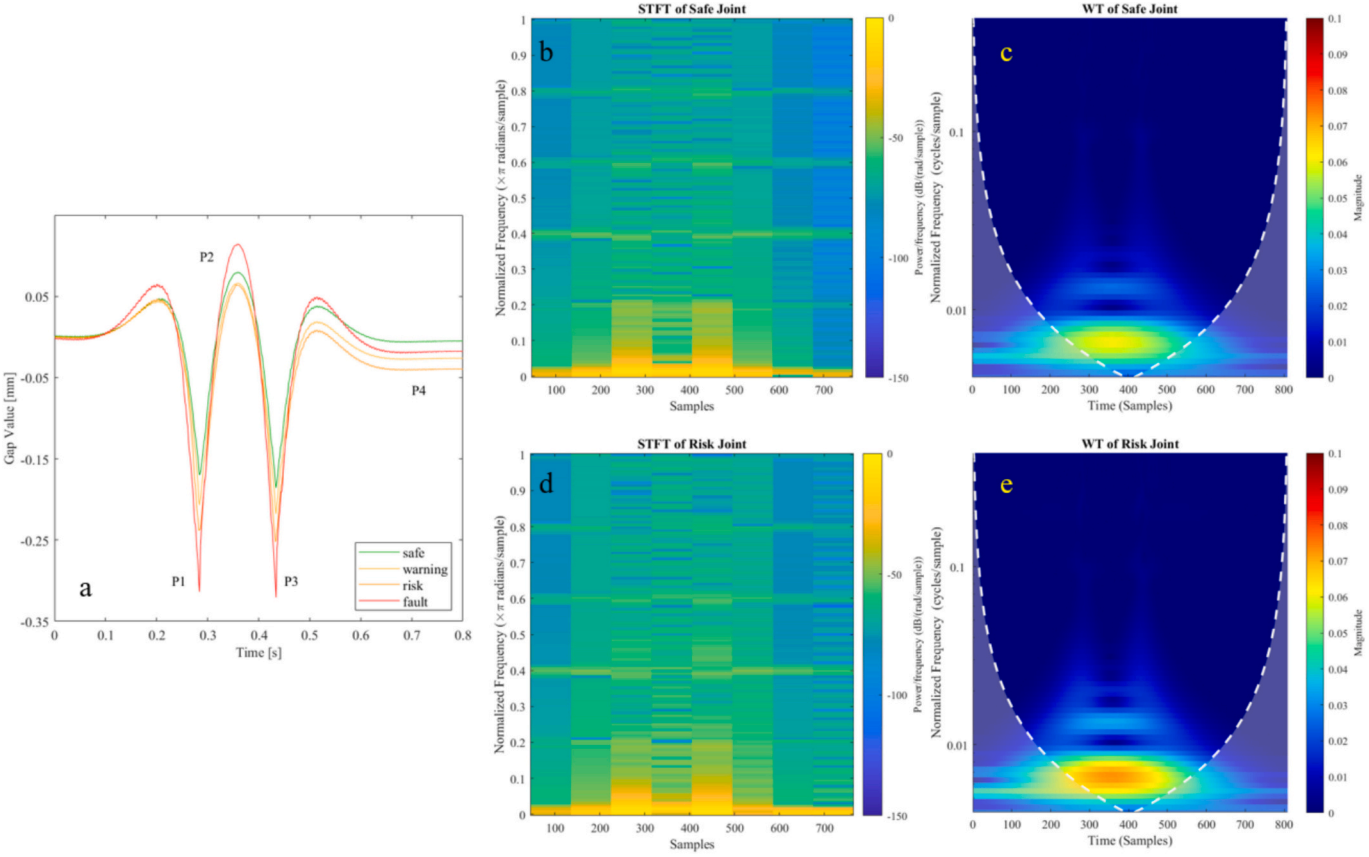


Fig. 5. **a** Examples of gap value for the 4 structural health classes considering a wheel-load of 60 kN; **b-e** spectrograms and scalograms for safe and risk joints.

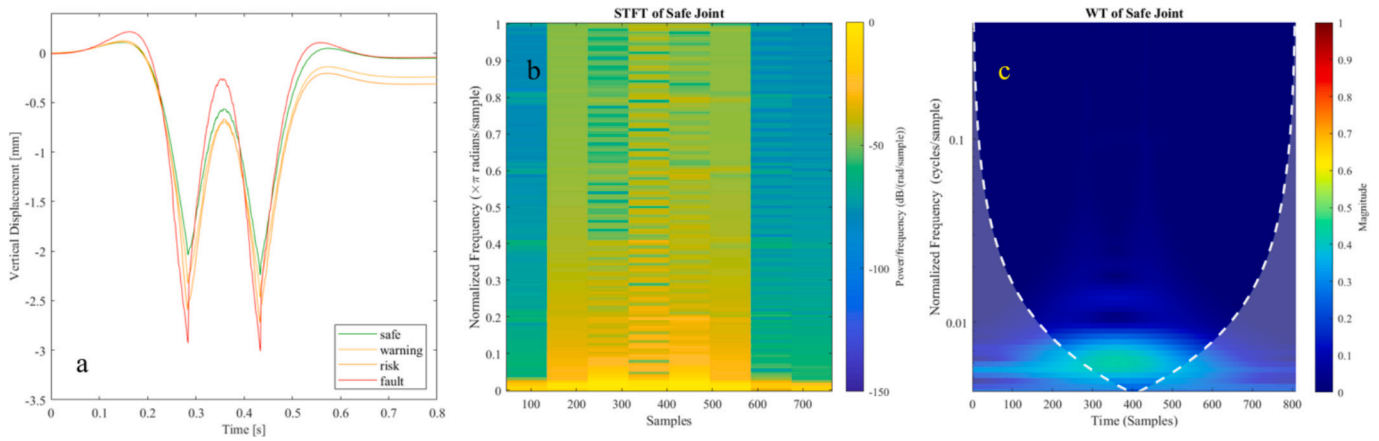


Fig. 6. **a** Examples of vertical displacement for the 4 structural health classes considering a wheel-load of 60 kN; **b-c** spectrogram and scalogram for a safe joint.

Table 2
a–c Gap value and vertical displacement data, at significant positions of the wheels for the 4 structural health classes, considering a wheel-load of **a** 60 kN, **b** 90 kN, **c** 120 kN.

Structural health classes	Gap value [mm]				Vertical displacement [mm]			
	P1	P2	P3	P4	P1	P2	P3	P4
Safe	−0.170	0.079	−0.186	−0.006	−2.040	−0.567	−2.241	−0.054
Warning	−0.207	0.066	−0.218	−0.027	−2.326	−0.695	−2.452	−0.242
Risk	−0.237	0.064	−0.251	−0.040	−2.528	−0.671	−2.646	−0.313
Fault	−0.314	0.114	−0.320	−0.019	−2.926	−0.264	−3.008	−0.047

Structural health classes	Gap value [mm]				Vertical displacement [mm]			
	P1	P2	P3	P4	P1	P2	P3	P4
Safe	−0.352	0.063	−0.351	−0.074	−3.600	−1.163	−3.717	−0.429
Warning	−0.368	0.063	−0.365	−0.081	−3.654	−1.169	−3.771	−0.487
Risk	−0.406	0.109	−0.401	−0.065	−3.952	−0.743	−3.994	−0.295
Fault	−0.467	0.177	−0.456	−0.028	−4.232	−0.329	−4.232	−0.048

Structural health classes	Gap value [mm]				Vertical displacement [mm]			
	P1	P2	P3	P4	P1	P2	P3	P4
Safe	−0.452	0.090	−0.447	−0.095	−4.751	−1.528	−4.790	−0.555
Warning	−0.473	0.104	−0.469	−0.099	−4.925	−1.409	−5.009	−0.652
Risk	−0.534	0.175	−0.515	−0.062	−5.199	−0.771	−5.189	−0.213
Fault	−0.563	0.259	−0.554	−0.024	−5.259	−0.328	−5.257	−0.005

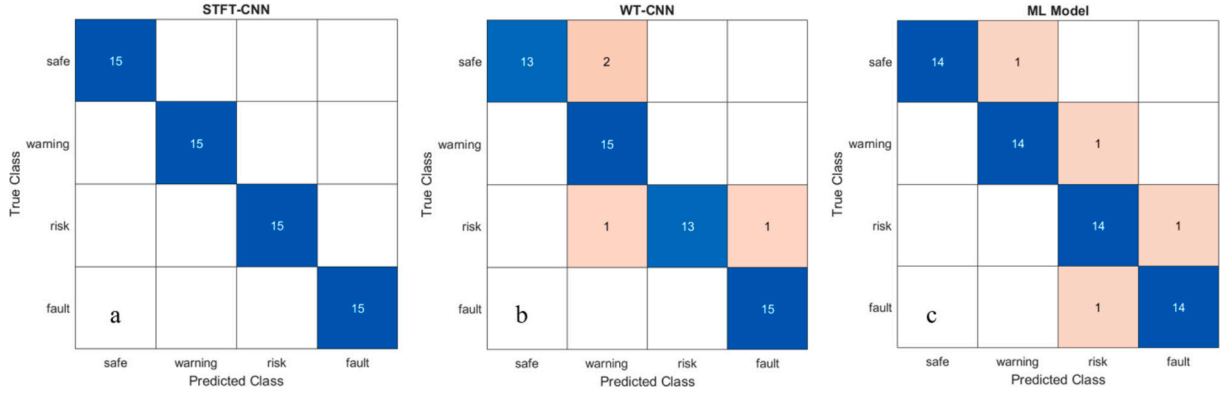


Fig. 7. Confusion Matrix on Test Data of gap value for **a** STFT-CNN, **b** WT-CNN and **c** ML classifier.

Residual misclassifications occur exclusively between adjacent health states (safe–warning and risk–fault), with no confusion between mechanically distant conditions. For vertical displacement signals (Fig. 8), confusion primarily arises from the limited structural influence

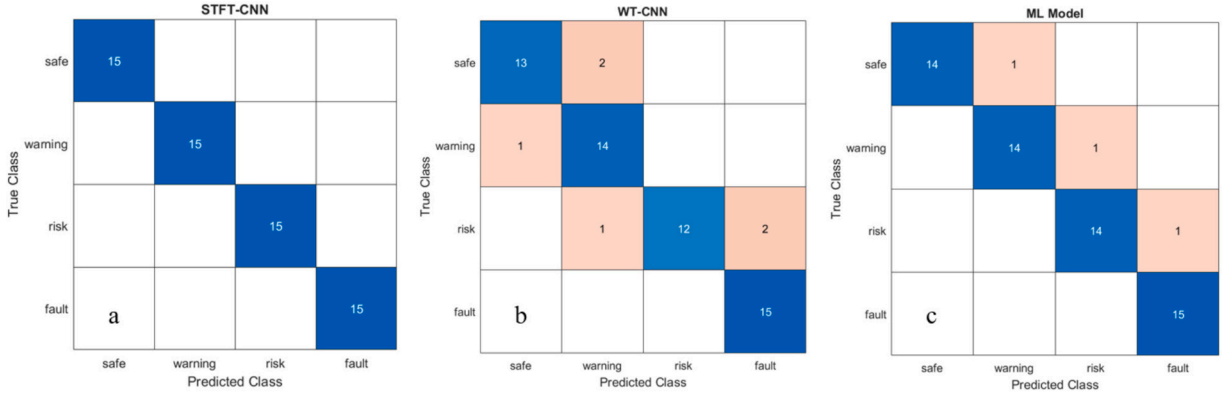


Fig. 8. Confusion Matrix on Test Data of vertical displacement for **a** STFT-CNN, **b** WT-CNN and **c** ML classifier.

Table 3
a–b Testing performance indicators for classifiers based on **a** gap value, **b** vertical displacement data.

Structural health classes	STFT-CNN			WT-CNN			ML		
	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)
Safe	100	100	100	100	86.7	92.9	100	93.3	96.6
Warning	100	100	100	83.3	100	90.9	93.3	93.3	93.3
Risk	100	100	100	100	86.7	92.9	87.5	93.3	90.3
Fault	100	100	100	93.8	100	96.8	93.3	93.3	93.3
Accuracy (%)		100			93.3			93.3	

Structural health classes	STFT-CNN			WT-CNN			ML		
	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)
Safe	100	100	100	92.9	86.7	89.7	100	93.3	96.6
Warning	100	100	100	82.4	93.3	87.5	93.3	93.3	93.3
Risk	100	100	100	100	80.0	88.9	93.3	93.3	93.3
Fault	100	100	100	88.2	100	93.8	93.8	100	96.8
Accuracy (%)		100			90.0			95.0	

of outer bolts. Partial loosening (warning class) induces only marginal amplification of quasi-static peaks associated with wheel–joint interaction (P1 and P3), producing time–frequency energy distributions closely resembling those of the safe condition. Similarly, the risk-to-fault transition mainly involves amplitude scaling rather than qualitative waveform changes. Gap value signals (Fig. 7) show comparable trends, where separability is governed by monotonic variations in joint flexibility and rail displacement.

Most misclassifications appear above the main diagonal, indicating a conservative tendency to overestimate degradation severity. Practically, a safe joint classified as warning would trigger increased inspection, whereas underestimation of severe damage is rare, and no fault condition is misclassified as safe. This bounded error structure aligns with predictive maintenance priorities, where avoiding missed critical defects outweighs occasional false alarms.

ROC-curve analysis supports these observations: the STFT-CNN achieves ideal performance (AUC = 1) for both indicators (Figs. 9–10), while other models yield AUC values near unity, with

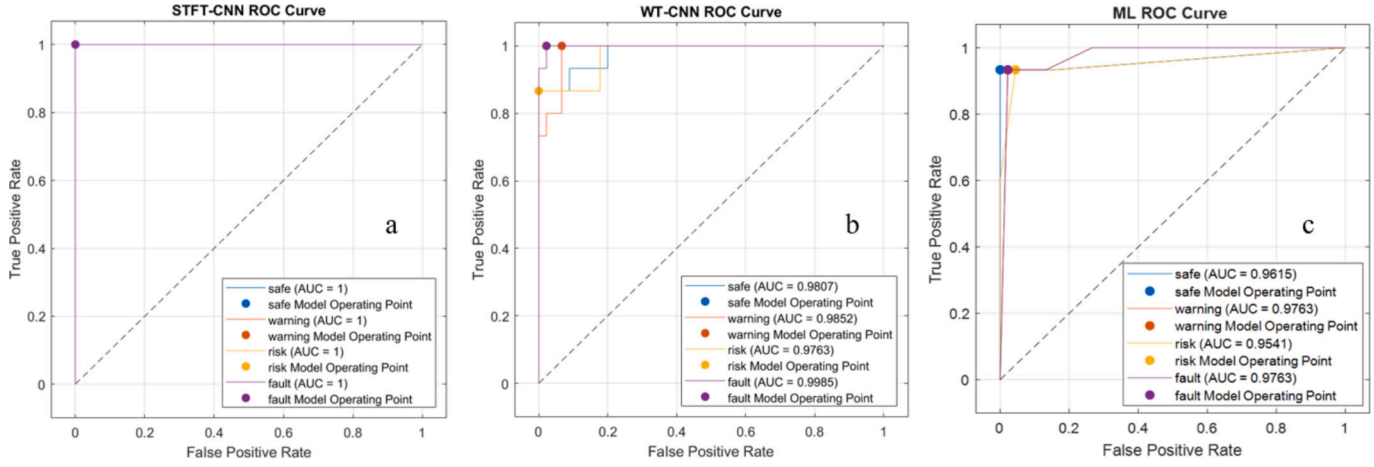


Fig. 9. ROC curves for the tested **a** STFT-CNN, **b** WT-CNN and **c** ML classifier based on gap value data.

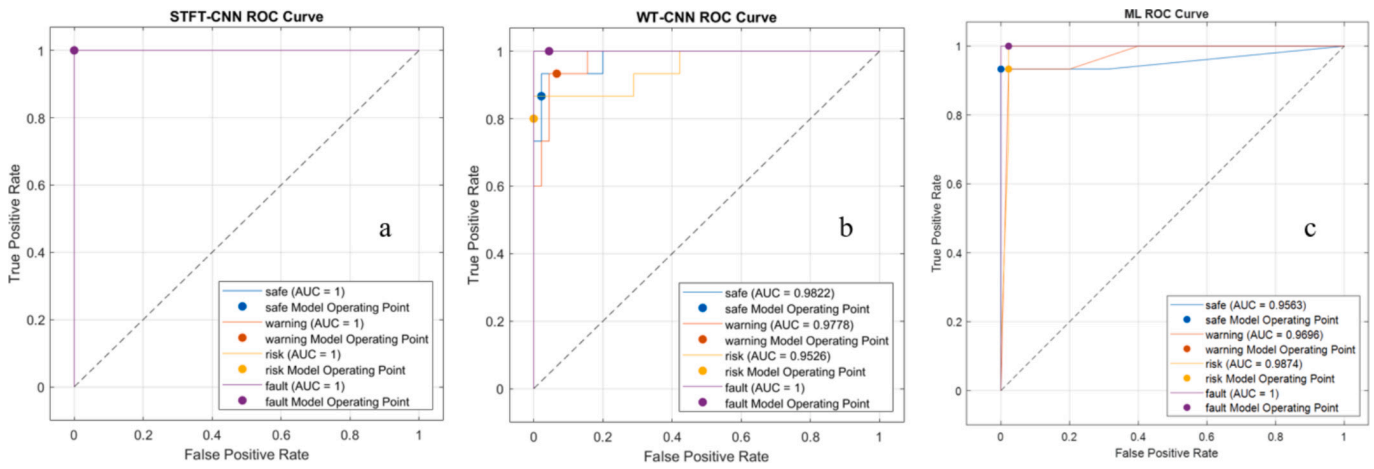


Fig. 10. ROC curves for the tested **a** STFT-CNN, **b** WT-CNN and **c** ML classifier based on vertical displacement data.

operating points concentrated near (0,1). The fault class consistently lies closest to this optimal region, demonstrating robust identification of critical conditions. These results highlight the superior diagnostic capability of DL models, which autonomously extract discriminative features, while traditional ML approaches require more extensive feature engineering [16].

3.1.3. *Misclassification analysis and physical interpretation*

A qualitative analysis of classification errors confirms that misclassifications arise from physically meaningful sources rather than algorithmic deficiencies. Adjacent-class errors reflect the subtle impact of partial bolt loosening and amplitude differences between neighbouring degradation levels. The Digital Twin generates mechanically ordered degradation regimes, enabling classifiers to establish physically interpretable decision boundaries.

From a safety perspective, the error structure is predominantly conservative, with rare underestimation of severe damage. False positives (overestimating degradation) trigger precautionary inspection, while false negatives (underestimating damage) are minimal. This conservative bias ensures the framework's suitability for automated predictive maintenance in safety-critical infrastructure, maintaining a clear prioritization of operational safety over occasional inspection redundancy.

3.1.4. *Transferability and robustness under real-world uncertainty*

The near-perfect classification performance reported in Sections 3.1.2–3.1.3 is an upper-bound benchmark, reflecting mechanically consistent, noise-free Digital Twin simulations. Real-world variability, including sensor noise, placement tolerances, material heterogeneity, environmental effects, and vehicle–track interaction fluctuations, may partially reduce feature separability and increase misclassification rates, particularly for adjacent degradation states (e.g., safe vs. early-warning).

Nevertheless, the primary degradation mechanisms, bolt preload loss and associated stiffness reduction, are well-established drivers of IRJ deterioration and are accurately reproduced by the validated Digital Twin. Prior experimental studies [16] demonstrate that classifiers trained on such synthetic data retain predictive relevance under real measurements, providing initial evidence of transferability.

The Digital Twin thus serves as a structured baseline for physically interpretable, systematically labelled training, enabling robust feature learning before exposure to operational uncertainties. Future developments will integrate laboratory and field measurements via hybrid virtual–physical updating and transfer-learning strategies, progressively introducing noise and environmental variability. This staged approach is expected to enhance model generalization while preserving mechanical interpretability.

Accordingly, the results presented in this section demonstrate intrinsic separability of degradation states under controlled conditions, providing a reproducible benchmark for subsequent field validation and operational deployment.

3.2. *Identification of bolt-loosening configurations within the warning class*

This case study extends the analysis to finer-grained diagnostics by evaluating the ability of the framework to discriminate among different bolt-loosening configurations within the same degradation class.

3.2.1. *Predictive objective and dataset preparation*

The model targets a finer-grained diagnosis within the warning class of bonded IRJs, defined by a reduction of at least one bolt pair to 66.6% of the nominal preload. Digital-twin simulations under a constant wheel load of 60 kN produced a balanced dataset of 150 samples, using gap values and vertical displacements near the IRJ as predictive indicators. Five warning subclasses were considered for supervised learning, corresponding to distinct bolt-loosening configurations: outer bolts (B14), central bolts (B23), an alternating pair (B24), bolts connecting the first rail (B12), and all bolts loosened (B1234). This approach extends the analysis from global stiffness degradation (Section 3.1.2) to the identification of specific loosening patterns in the warning category.

As shown in Figs. S1–S2 of the Supplementary Material and summarized in Table 4, the gap and displacement responses are largely consistent across these configurations, confirming that different loosening patterns can produce similar mechanical signatures. This supports their aggregation into a single health category for standard maintenance decisions. An exception is the B14 configuration, which systematically yields lower gap and displacement values, approaching those of the safe condition. This explains the occasional misclassifications observed in Figs. 7b–8b and highlights the limited structural role of the outer bolts.

Relative to the broader classification in Section 3.1, where 90 samples were sufficient to characterize the warning class across multiple wheel loads, the present subclass-level analysis required 150 samples under a single load condition. While not generalizable, this increase illustrates the higher data demand associated with finer diagnostic resolution, as required for identifying specific bolt-loosening configurations.

3.2.2. *Evaluating predictive performance*

The validation confusion matrices for the STFT-CNN, WT-CNN, and ML decision tree classifiers are reported in Figs. S3 and S4 of the Supplementary Material for models trained on gap values and vertical displacements, respectively, with corresponding performance metrics

Table 4
Gap value and vertical displacement data, at significant positions of the wheels for the 5 loosened bolt configurations, considering a wheel-load of 60 kN.

Bolt configurations	Gap value [mm]				Vertical displacement [mm]			
	P1	P2	P3	P4	P1	P2	P3	P4
B14	−0.175	0.080	−0.191	−0.006	−2.072	−0.560	−2.270	−0.057
B24	−0.207	0.066	−0.218	−0.027	−2.326	−0.695	−2.452	−0.242
B12	−0.218	0.063	−0.222	−0.031	−2.448	−0.725	−2.510	−0.277
B23	−0.229	0.051	−0.235	−0.043	−2.411	−0.724	−2.510	−0.262
B1234	−0.235	0.052	−0.241	−0.044	−2.443	−0.725	−2.550	−0.277

summarized in Table 5a and b. Overall, both DL- and ML-based approaches achieve high predictive accuracy (≈90%) for both input types, confirming the suitability of the selected indicators for maintenance-oriented diagnostics.

Error analysis reveals partial confusion between the B14 and B24 configurations, as well as between B23 and B1234, in the WT-CNN and ML models. This behaviour is consistent with the dominant structural role of the central bolts (B2 and B3), which largely govern IRJ stiffness. Notably, only the STFT-CNN consistently resolves these closely related configurations, highlighting its superior discriminative capability. ROC analysis (not shown for brevity) further supports this result: the STFT-CNN attains ideal performance (AUC = 1), whereas the ML classifier exhibits its lowest AUC (0.82) for the B24 subclass.

As discussed in Section 3.1, different bolt-loosening patterns within the same health class produce highly similar gap values and displacement signatures, naturally steering classifiers toward identifying the global joint condition rather than individual bolt configurations. Section 3.2 demonstrates that finer diagnostic resolution can nevertheless be achieved through targeted training on expanded, class-specific datasets. While this strategy could be extended to other critical joint states, identifying every possible loose-bolt combination is not essential for the objectives of this study. Such an approach would substantially increase the number of predictive classes and training requirements, leading to limited practical benefit relative to the added computational complexity.

3.3. Combined bolt loosening and adjacent sleeper degradation assessment

This case study examines the robustness of the proposed methodology under coupled degradation mechanisms involving both bolt-preload loss and adjacent sleeper stiffness reduction.

3.3.1. Predictive objective and dataset preparation

The model aims to predict the structural health of infrastructure sections subjected to various loose-bolt configurations and potential stiffness loss in the sleeper adjacent to the bonded IRJ under two wheel-loads (60 and 90 kN). Digital-twin simulations yielded a balanced dataset of 600 samples using gap values and vertical displacements near the IRJ as predictive indicators. Six structural health classes were defined for supervised learning: safe (all bolts preloaded at 75 kN), B14 and B23 (outer and central bolt pairs loosened to 25 kN, respectively), S16 (all bolts intact, sleeper 16 adjacent to the joint exhibiting a 50% stiffness reduction to 0.0065 N/mm^3), and S16-B14 and S16-B23, which combine sleeper degradation with the corresponding bolt-loosening defects.

Representative gap and vertical displacement responses for the six classes under a 60 kN wheel load are shown in Figs. S5a and S6a of the Supplementary Material, while Figs. S5b–c and S6b–c present the associated time–frequency representations for the B23 and S16-B23 cases. Numerical values at four critical wheel positions (P1–P4; see Section 3.1.2) are reported in Table 6a–b and were used as inputs for ML training. Across the considered scenarios, the S16 configuration exhibits response levels comparable to those of the B14 and B23 classes, with the main divergence observed in vertical displacement under the higher wheel load (90 kN). This result confirms that a localized reduction in sleeper stiffness can influence the structural response to a degree comparable to bolt-loosening defects, underscoring the need to jointly monitor fastener integrity and support conditions in IRJ health assessment.

3.3.2. Evaluating predictive performance

The validation confusion matrices for the STFT-CNN, WT-CNN, and ML decision tree classifiers are reported in Figs. S7 and S8 of the Supplementary Material for models trained on gap values and vertical displacements, respectively, with corresponding performance metrics summarized in Table 7a–b. All approaches achieve predictive accuracies

Table 5
a–b Testing performance indicators for classifiers based on **a** gap value, **b** vertical displacement data.

Structural health classes	STFT-CNN			WT-CNN			ML		
	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)
B14	100	100	100	75.0	100	85.7	75.0	100	85.7
B24	100	100	100	100	66.7	80.0	100	66.7	90.0
B12	100	100	100	100	100	100	100	100	100
B23	100	100	100	100	83.3	90.9	100	66.7	80.0
B1234	100	100	100	85.7	100	92.3	75.0	100	85.7
Accuracy (%)		100			90.0			86.7	

Structural health classes	STFT-CNN			WT-CNN			ML		
	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)
B14	100	100	100	100	100	100	75.0	100	85.7
B24	100	100	100	100	100	100	100	66.7	80.0
B12	100	100	100	100	100	100	83.3	83.3	83.3
B23	100	100	100	100	66.7	80.0	100	100	100
B1234	100	100	100	75.0	100	85.7	83.3	83.3	83.3
Accuracy (%)		100			93.3			86.7	

Table 6
a–b Gap value and vertical displacement data, at significant positions of the wheels for the 6 structural health classes, considering a wheel-load of a 60 kN, b 90 kN.

Structural health classes	Gap value [mm]				Vertical displacement [mm]			
	P1	P2	P3	P4	P1	P2	P3	P4
Safe	−0.170	0.079	−0.186	−0.006	−2.040	−0.567	−2.241	−0.054
B14	−0.232	0.046	−0.239	−0.045	−2.528	−0.781	−2.573	−0.300
B23	−0.274	0.059	−0.272	−0.054	−2.641	−0.649	−2.677	−0.298
S16	−0.212	0.046	−0.258	−0.025	−2.774	−1.051	−3.261	−0.139
S16-B23	−0.336	0.015	−0.348	−0.065	−3.729	−1.189	−3.819	−0.389
S16-B14	−0.362	−0.016	−0.351	−0.095	−3.940	−1.509	−3.906	−0.663

Structural health classes	Gap value [mm]				Vertical displacement [mm]			
	P1	P2	P3	P4	P1	P2	P3	P4
Safe	−0.352	0.063	−0.351	−0.074	−3.600	−1.162	−3.717	−0.429
B14	−0.420	0.109	−0.405	−0.066	−4.121	−1.045	−4.152	−0.583
B23	−0.362	0.124	−0.355	−0.045	−3.727	−0.773	−3.727	−0.250
S16	−0.447	0.000	−0.466	−0.098	−5.133	−2.015	−5.366	−0.614
S16-B23	−0.487	0.058	−0.476	−0.059	−5.356	−1.534	−5.388	−0.313
S16-B14	−0.586	0.043	−0.561	−0.084	−6.255	−1.892	−6.220	−0.751

Table 7
a–b Testing performance indicators for classifiers based on **a** gap value, **b** vertical displacement data.

Structural health classes	STFT-CNN			WT-CNN			ML		
	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)
Safe	100	100	100	90.9	100	95.2	100	100	100
B14	100	100	100	100	90.0	94.7	94.7	90.0	92.3
B23	100	100	100	86.4	95.0	90.5	90.5	95.0	92.7
S16	100	100	100	100	100	100	95.2	100	97.7
S16-B23	100	100	100	100	95.0	97.4	100	95.0	97.4
S16-B14	100	100	100	100	95.0	97.4	95.0	95.0	95.0
Accuracy (%)		100			95.8			95.8	

Structural health classes	STFT-CNN			WT-CNN			ML		
	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)
Safe	100	100	100	95.2	100	97.6	95.2	100	97.6
B14	90.9	100	95.2	90.9	100	95.2	95.2	100	97.6
B23	100	95.0	97.4	100	95.0	97.4	100	100	100
S16	100	100	100	100	100	100	95.2	100	97.6
S16-B23	100	100	100	100	95.0	97.4	100	90.0	94.7
S16-B14	100	95.0	97.4	100	95.0	97.4	100	95.0	97.4
Accuracy (%)		98.3			97.5			97.5	

above 90% for both input features, confirming the robustness and practical relevance of the proposed diagnostic framework.

Inspection of the misclassification patterns (Figs. S7 and S8) reveals no systematic anomalies. The most frequent confusions occur between the bolt-loosening configurations B23 and B14, and between B14 and the combined defect S16-B14, as discussed in Section 3.2. These errors indicate that the effects of sleeper stiffness loss can occasionally be masked by bolt-related signatures. Nevertheless, the consistently high overall accuracy suggests that such ambiguities are likely resolved through repeated measurements over time, enhancing the reliability of the framework for maintenance decision-making.

ROC analysis (omitted for brevity) corroborates these observations. The STFT-CNN achieves near-ideal performance, with an AUC approaching unity, while all models exhibit operating points clustered near the optimal (0,1) region, indicating strong discriminative capability. Overall, the results confirm the superiority of DL-based approaches for fault diagnosis, as they autonomously extract complex patterns from raw data and achieve high accuracy with minimal feature engineering, outperforming traditional ML models that require more extensive preprocessing and tuning to reach comparable performance.

3.4. Detection of sleeper stiffness loss near the bonded IRJ

This case study focuses on the detection and classification of localized sleeper-support degradation in the vicinity of the bonded IRJ under variable loading conditions.

3.4.1. Predictive objective and dataset preparation

The model investigates the detection of sleeper stiffness loss in the vicinity of a bonded IRJ under two wheel-loads (60 and 90 kN). Digital-twin simulations produced a balanced dataset of 600 samples using gap values and vertical displacements near the joint as predictive indicators. All bolts are assumed to be correctly preloaded, isolating the effect of sleeper degradation. Worn sleepers are modelled through a 50% reduction in stiffness per unit area (0.0065 N/mm^3), relative to the nominal value of 0.013 N/mm^3, corresponding to a typical track stiffness of 21 N/mm^2.

Six structural health classes were defined for supervised learning: safe (all sleepers in standard condition); S15 and S16 (stiffness loss in the sleepers immediately adjacent to the joint on the first and second rail sections, respectively); S15–S16 (simultaneous degradation of both adjacent sleepers); and S8 and S23 (damaged sleepers located

approximately 4.5 m from the joint, at the midpoints of the first and second rail sections, respectively, each section spanning about 9 m in the DT).

Representative gap and vertical displacement responses for the six classes under a 60 kN wheel load are shown in Figs. S9a and S10a of the Supplementary Material, with corresponding time–frequency representations reported in Figs. S9b–c and S10b–c. Numerical values at four critical wheel positions (P1–P4; see Section 3.1.2) are summarized in Table 8a–b and used as inputs for ML training. The results indicate that the safe, S8, and S23 configurations exhibit similar responses, implying that stiffness loss in sleepers located farther from the joint is difficult to detect. In contrast, degradation in sleepers adjacent to the IRJ (S15 and S16) produces markedly stronger signatures, with the combined S15–S16 case representing the most critical condition.

3.4.2. Evaluating predictive performance

The validation confusion matrices for the STFT-CNN, WT-CNN, and ML decision tree classifiers are reported in Figs. S11 and S12 of the Supplementary Material for models trained on gap values and vertical displacements, respectively, with corresponding performance metrics summarized in Table 9a–b. Overall, all models demonstrate satisfactory classification capability, with accuracies ranging from 82% for the WT-CNN using gap values to above 90% for the ML classifiers, and perfect performance for the STFT-CNN.

Both predictive indicators prove effective for identifying maintenance-relevant sleeper degradation near the bonded IRJ. Nevertheless, strong similarities among certain classes lead to recurring misclassifications. For example, the WT-CNN trained on gap values achieves a recall of only 24% for the safe class, frequently misclassifying it as S23. This behaviour indicates a conservative bias toward more critical conditions, which, although increasing false positives, may be acceptable in a PdM context. Similarly, the WT-CNN trained on vertical displacement data exhibits confusion between S8 and S23, reflecting the inherent difficulty of distinguishing stiffness loss located before or after the joint when their mechanical responses are nearly indistinguishable. In contrast, the STFT-CNN achieves flawless classification on the validation set.

ROC analysis corroborates these findings. The STFT-CNN attains ideal performance with an AUC of 1 for both gap value and vertical displacement inputs (Figs. 11–12), while most other models also exhibit AUC values close to unity. An exception is the WT-CNN trained on gap values, for which the safe class operating point lies at (0.02, 0.15), indicating limited sensitivity to healthy conditions.

The comparatively strong performance of the ML classifier relative to the WT-CNN is attributable to its training on four strategically selected wheel positions (P1–P4) that concentrate the most informative structural features. While this approach requires careful pre-processing and expert validation, it enhances discrimination among similar classes. By contrast, DL models such as the STFT-CNN operate directly on raw signals, enabling fully automated analysis. In this case study, however, the WT-CNN proved less effective at extracting sufficiently discriminative patterns, underscoring the importance of representation choice in DL-based PdM applications.

3.5. Local detection of remote sleeper degradation using sparse displacement measurements

This case study evaluates the feasibility of detecting remote sleeper degradation using sparse local displacement measurements, with particular emphasis on sensor-location sensitivity.

3.5.1. Predictive objective and dataset preparation

The model evaluates stiffness degradation of Sleeper 8, located approximately 4.5 m from the bonded IRJ in the first rail section, under wheel loads of 60 and 90 kN. A balanced dataset of 300 samples was generated from 100 DT simulations, each providing vertical displacement measurements at sleepers 8, 11, and 14 (p8, p11, p14). With a sleeper pitch of 0.6 m, these positions are spaced 1.8 m apart, allowing independent assessment of how sensor location affects classification performance. Two structural health classes were defined: safe, representing nominal track behaviour with a stiffness per area of 0.013 N/mm^3 (21 N/mm^2 track stiffness), and S8, representing a 50% stiffness reduction (0.0065 N/mm^3). All bonded IRJs were assumed to have fully preloaded bolts.

Vertical displacement responses under both wheel loads are shown in Figs. S13a and S14a of the Supplementary Material, with corresponding time–frequency representations at p8 provided in Figs. S13b–c and S14b–c. Table 10 summarizes representative displacement values at four key wheel positions: the two lower peaks during wheel passages (P1 and P3), the intermediate upper peak (P2), and the residual displacement after both wheels have passed (P4). Notably, P4 exhibits near-zero displacement across all configurations, offering minimal discriminatory value. As expected, the contrast between safe and S8 is largest at p8, where the stiffness reduction occurs, and diminishes progressively at p11 and p14 as the measurement location moves away from the degraded sleeper, highlighting the influence of sensor placement on predictive accuracy.

3.5.2. Evaluating predictive performance

The validation confusion matrices for the STFT-CNN, WT-CNN, and

Table 8
a–b Gap value and vertical displacement data, at significant positions of the wheels for the 6 structural health classes, considering a wheel-load of **a** 60 kN, **b** 90 kN.

Structural health classes	Gap value [mm]				Vertical displacement [mm]			
	P1	P2	P3	P4	P1	P2	P3	P4
Safe	−0.170	0.079	−0.186	−0.006	−2.040	−0.567	−2.241	−0.054
S8	−0.186	0.078	−0.189	−0.007	−2.273	−0.612	−2.297	−0.057
S15	−0.245	0.040	−0.274	−0.034	−3.180	−1.187	−3.525	−0.339
S16	−0.212	0.046	−0.258	−0.025	−2.774	−1.051	−3.261	−0.139
S23	−0.171	0.080	−0.188	−0.007	−2.077	−0.602	−2.293	−0.053
S15-S16	−0.487	−0.116	−0.490	−0.120	−6.146	−2.916	−6.255	−0.945

Structural health classes	Gap value [mm]				Vertical displacement [mm]			
	P1	P2	P3	P4	P1	P2	P3	P4
Safe	−0.352	0.063	−0.351	−0.074	−3.600	−1.162	−3.717	−0.429
S8	−0.348	0.064	−0.354	−0.074	−3.621	−1.218	−3.798	−0.436
S15	−0.465	−0.003	−0.459	−0.099	−5.316	−2.156	−5.341	−0.714
S16	−0.447	0.000	−0.466	−0.098	−5.133	−2.015	−5.366	−0.614
S23	−0.354	0.064	−0.355	−0.076	−3.675	−1.222	−3.799	−0.434
S15-S16	−0.712	−0.201	−0.716	−0.204	−9.503	−4.811	−9.518	−1.823

Table 9
a–b Testing performance indicators for classifiers based on **a** gap value, **b** vertical displacement data.

Structural health classes	STFT-CNN			WT-CNN			ML		
	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)
Safe	100	100	100	60.0	15.0	24.0	90.5	95.0	92.7
S8	100	100	100	100	100	100	95.0	95.0	95.0
S15	100	100	100	90.9	100	95.2	100	100	100
S16	100	100	100	100	90.0	94.7	100	100	100
S23	100	100	100	51.4	90.0	65.5	100	95.0	97.4
S15-S16	100	100	100	100	100	100	100	100	100
Accuracy (%)		100			82.5			97.5	

Structural health classes	STFT-CNN			WT-CNN			ML		
	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)
Safe	100	100	100	94.1	80.0	86.5	86.4	95.0	90.5
S8	100	100	100	80.0	100	88.9	88.9	80.0	84.2
S15	100	100	100	80.0	100	88.9	100	100	100
S16	100	100	100	100	90.0	94.7	100	100	100
S23	100	100	100	92.3	60.0	72.7	85.0	85.0	85.0
S15-S16	100	100	100	90.9	100	95.2	100	100	100
Accuracy (%)		100			88.3			93.3	

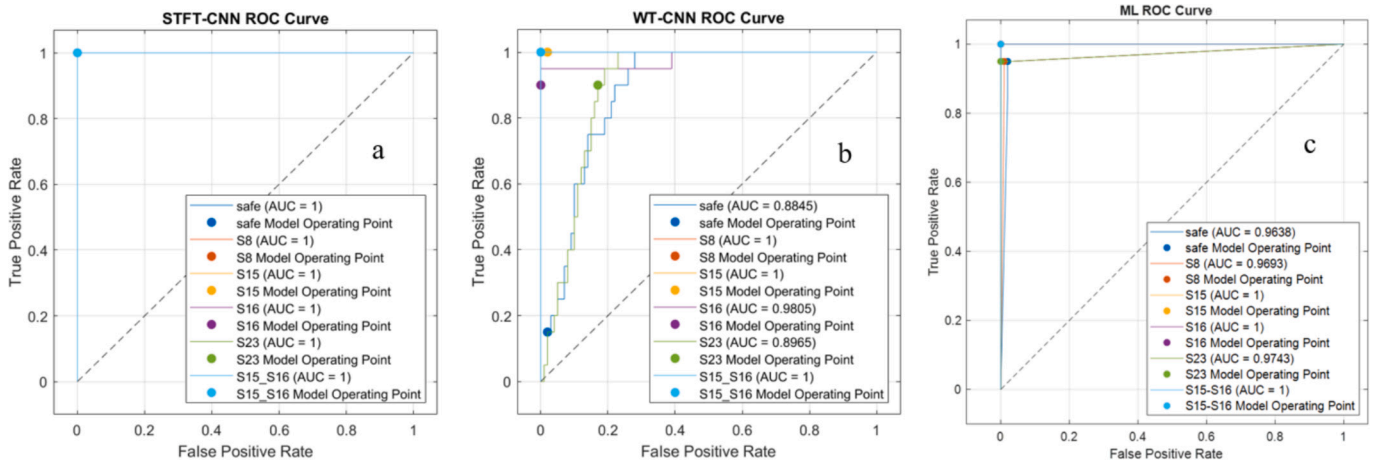


Fig. 11. ROC curves for the tested **a** STFT-CNN, **b** WT-CNN and **c** ML classifier based on gap value data.

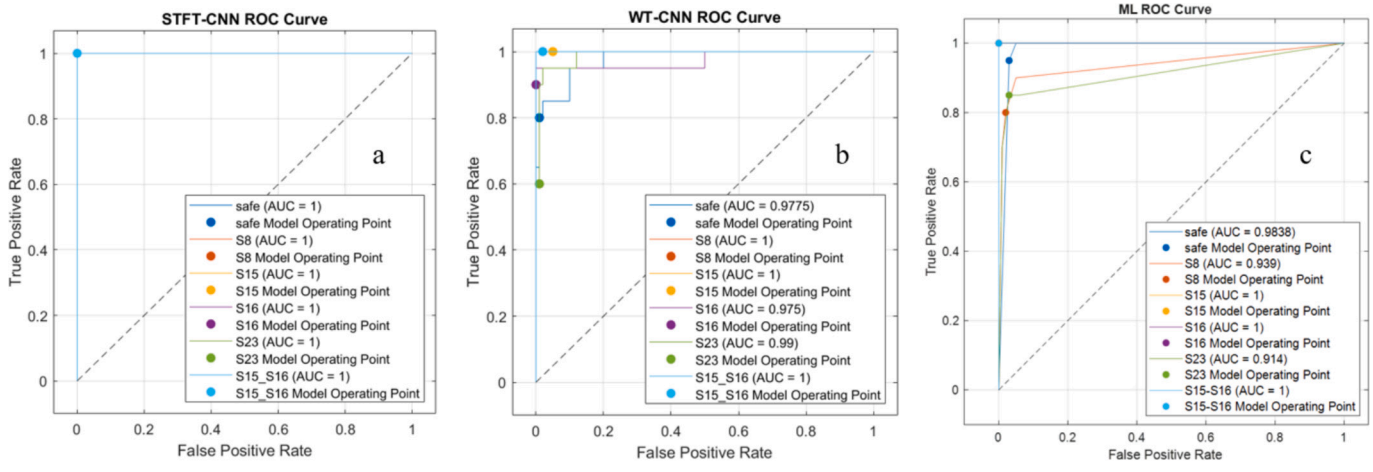


Fig. 12. ROC curves for the tested **a** STFT-CNN, **b** WT-CNN and **c** ML classifier based on vertical displacement data.

ML decision-tree classifiers are shown in Figs. S15 and S16 of the Supplementary Material, corresponding to models trained and tested on vertical displacement measurements acquired near sleepers 8 and 14. Performance metrics are summarized in Table 11a–c. Both the STFT-CNN and ML classifiers maintain high accuracy even as the measurement location shifts away from the degraded sleeper. A monitoring

Table 10
Vertical displacement data, at significant positions of the wheels for safe and S8 predictive classes, different measurement points, p8, p11 and p14, considering two wheel-loads of 60 kN and 90 kN.

Structural health classes	Vertical displacement [mm] – wheel-load 60 kN				Vertical displacement [mm] – wheel-load 90 kN			
	P1	P2	P3	P4	P1	P2	P3	P4
Safe-p8	−1.536	−0.770	−1.533	0.000	−2.304	−1.155	−2.299	−0.003
S8-p8	−2.313	−1.193	−2.308	−0.001	−3.470	−1.789	−3.462	−0.004
Safe-p11	−1.537	−0.781	−1.504	0.003	−2.305	−1.166	−2.221	0.026
S8-p11	−1.613	−0.869	−1.544	0.004	−2.420	−1.296	−2.286	0.028
Safe-p14	−1.488	−0.750	−1.548	0.001	−2.311	−1.096	−2.297	0.028
S8-p14	−1.474	−0.786	−1.587	0.002	−2.340	−1.175	−2.358	0.027

Table 11
a–c Testing performance indicators for classifiers based on vertical displacement measured at **a** sleeper 8, **b** sleeper 11, **c** sleeper 14.

Structural health classes	STFT-CNN			WT-CNN			ML		
	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)
Safe-p8	100	100	100	76.9	100	87.0	100	90.0	94.7
S8-p8	100	100	100	100	70.0	82.4	90.9	100	95.2
Accuracy (%)		100			85			95.0	

Structural health classes	STFT-CNN			WT-CNN			ML		
	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)
Safe-p11	100	100	100	100	40.0	57.1	100	100	100
S8-p11	100	100	100	62.5	100	76.9	100	100	100
Accuracy (%)		100			70			100	

Structural health classes	STFT-CNN			WT-CNN			ML		
	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)
Safe-p14	100	100	100	66.7	100	80.0	100	100	100
S8-p14	100	100	100	100	50.0	66.7	100	100	100
Accuracy (%)		100			75			100	

distance of 3.6 m, from sleeper 8 to sleeper 14, was adopted as an initial benchmark for efficient SHM design. Given the large spatial extent of railway networks, future work could explore the maximum sensor spacing that preserves predictive reliability while minimizing instrumentation requirements. In contrast, the WT-CNN exhibits lower robustness, with accuracy dropping from ~85% at p8 to 70–75% at p11 and p14.

ROC analysis for measurements at sleeper 8 (Fig. 13) reinforces these findings: operating points cluster near the ideal (0,1) coordinate, producing AUC values close to unity across classifiers. As noted in Section 3.4, the ML classifier outperforms the WT-CNN because it leverages four carefully selected displacement points (P1–P4) with high diagnostic relevance. Deep learning models, in contrast, process the full waveform, enabling fully automated predictions but reducing emphasis on the most informative features. Nevertheless, DL classifiers offer rapid retraining and adaptability to new structural configurations, supporting scalable deployment in evolving SHM frameworks.

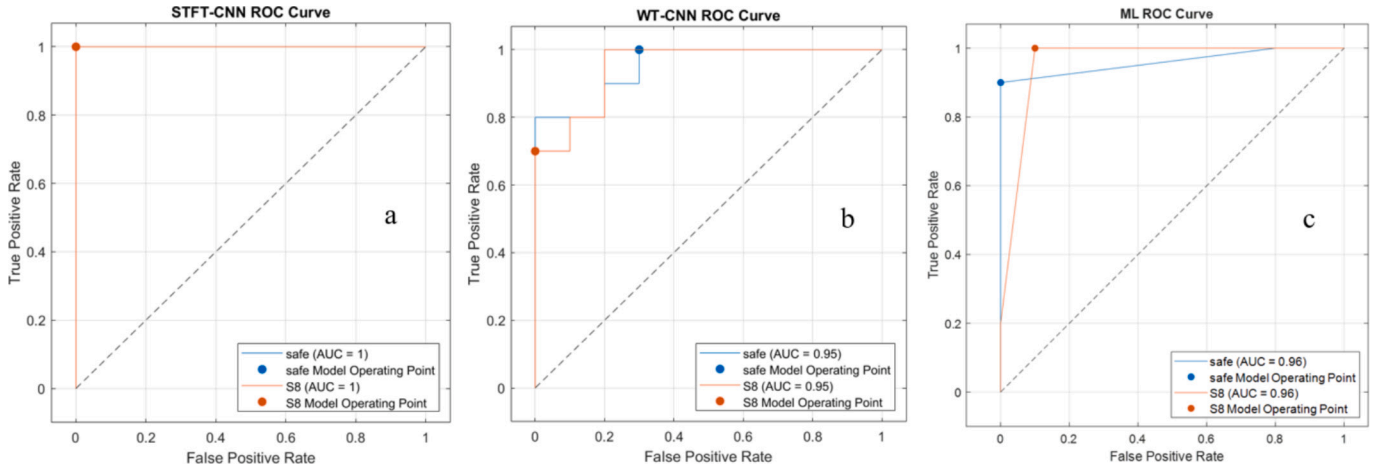


Fig. 13. ROC curves for the tested **a** STFT-CNN, **b** WT-CNN and **c** ML classifier based on vertical displacement measured at sleeper 8.

3.6. Full-field displacement profile analysis for remote sleeper degradation

This case study investigates the effectiveness of full-field displacement profiles for identifying remote sleeper degradation and assessing the influence of measurement timing on predictive performance.

3.6.1. Predictive objective and dataset preparation

The model evaluates stiffness degradation in Sleeper 8, located at the midpoint of the first rail section (~4.5 m from the bonded IRJ), under wheel loads of 60 and 90 kN. A balanced dataset of 500 samples was generated from 100 DT simulations, each providing five vertical displacement profiles along the full 18 m track segment (two 9 m rails joined by a bonded IRJ). Displacement fields were recorded at five time instants, t34, t134, t234, t284, and t358, corresponding to wheel positions relative to critical sleepers and the joint. These profiles, spaced approximately 0.1 s apart, were used independently to train and evaluate predictive models, enabling assessment of how measurement timing affects classification accuracy. Two structural health classes were defined: safe, representing nominal track behaviour with a stiffness per area of 0.013 N/mm^3 (21 N/mm^2 track stiffness), and S8, representing a 50% reduction to 0.0065 N/mm^3. All bonded IRJs were assumed to have fully preloaded bolts.

The acquisition times correspond to the position of the wheelset: t34, above Sleeper 8; t134, above Sleeper 11; t234, above Sleeper 14; t284, above the bonded IRJ; and t358, with the wheelset symmetrically positioned across the joint. Representative displacement profiles for both classes at t34 and t284 under 60 and 90 kN loads are shown in Figs. S17a and S18a, with corresponding time–frequency representations in Figs. S17b–c and S18b–c. Table 12 reports characteristic values at four key points of the displacement profile: the two lower peaks from wheel passages (P1 and P3), the intermediate upper peak (P2), and the displacement near the midpoint of the second rail (P4). As expected, when the wheel is directly above the degraded sleeper (t34), the displacement profiles for the safe and S8 conditions diverge markedly, enhancing class separability. As the wheel moves away from Sleeper 8, the profiles gradually converge, as evident near the IRJ (t284), highlighting the critical influence of measurement timing on diagnostic sensitivity.

3.6.2. Evaluating predictive performance

The validation confusion matrices for the STFT-CNN, WT-CNN, and ML decision-tree classifiers trained and tested using the vertical displacement profile at time instant t358 are shown in Fig. S19 of the Supplementary Material. Since the performance metrics of models trained at previous time points are summarized in Table 13a–c, additional images have been omitted.

The results show that all classifiers, both deep CNN architectures and the ML model, achieve excellent predictive accuracy, consistently exceeding 90% when using the full displacement profiles as input. For the WT-CNN and ML classifier, misclassification rates increase as the displacement field is sampled at times progressively farther from the wheel passage over the degraded sleeper, reflecting reduced class separability. In contrast, the STFT-CNN maintains stable, high performance across all sampling times.

ROC analysis (not shown to avoid redundancy) corroborates these findings, with all classifiers achieving an AUC of 1, indicating perfect classification. Overall, these results confirm the robustness of the proposed framework and highlight the practical effectiveness of vertical displacement profiles as predictive indicators for railway maintenance applications.

4. Discussion

This section interprets the results from the perspective of automation-oriented decision-making and PdM, rather than focusing exclusively on classification accuracy. While the high-performance metrics reported in Table 14 confirm the technical validity of the proposed models, the primary contribution lies in clarifying which degradation mechanisms are reliably detectable, under what structural and sensing conditions, and with what level of automation confidence. This distinction is essential for the design of deployable DT–enabled monitoring systems and aligns with the broader objective of AI-supported infrastructure management in railway engineering [18,26], as well as recent developments in physics-informed monitoring and time–frequency-based diagnostics [54,55].

Building upon our previous work [16], which combined a Digital Twin with a decision-tree classifier under simplified modelling assumptions, the present study advances both modelling fidelity and automation capability. Specifically, it incorporates IRJ–sleeper coupling effects and exploits full time-series responses transformed into time–frequency representations for DCNN diagnostics. Earlier FE investigations characterised joint mechanics under variable loads but did not integrate predictive models, while many AI-based monitoring studies lack physics-informed structural coupling. The proposed framework bridges this gap by unifying high-fidelity mechanical modelling and AI-driven classification within a single automated decision pipeline. In doing so, it moves beyond conceptual DT–AI architectures and demonstrates a reproducible workflow capable of translating physically interpretable degradation scenarios into actionable maintenance insights.

4.1. Automation-driven interpretation of predictive performance

Across all six case studies (Cases 3.1–3.6), both deep learning (STFT-CNN and WT-CNN) and traditional machine learning classifiers achieve predictive accuracies frequently exceeding 90%. However, from an automation perspective, global accuracy alone is insufficient to evaluate practical suitability. A deployable monitoring strategy must consider the mechanical consistency of errors and the operational implications of

Table 12
Vertical displacement data, at significant positions of the wheels for safe and S8 predictive classes, at several time instants, considering two wheel-loads of 60 kN and 90 kN.

Structural health classes	Vertical displacement [mm] – wheel-load 60 kN				Vertical displacement [mm] – wheel-load 90 kN			
	P1	P2	P3	P4	P1	P2	P3	P4
Safe-t34	−1.509	−0.771	−1.538	0.000	−2.264	−1.156	−2.307	0.000
S8-t34	−1.520	−0.961	−2.315	0.000	−2.280	−1.440	−3.472	0.000
Safe-t134	−1.551	−0.772	−1.539	−0.001	−2.326	−1.158	−2.308	−0.002
S8-t134	−1.829	−1.148	−1.615	−0.001	−2.743	−1.721	−2.422	−0.002
Safe-t234	−1.551	−0.783	−1.490	0.000	−2.325	−1.167	−2.313	0.004
S8-t234	−1.776	−0.816	−1.470	0.001	−2.670	−1.220	−2.343	0.005
Safe-t284	−1.503	−0.641	−2.047	0.017	−2.224	−0.946	−3.571	0.030
S8-t284	−1.534	−0.695	−2.286	0.023	−2.289	−0.994	−3.602	0.033
Safe-t358	−1.582	−0.566	−1.587	0.059	−2.315	−1.163	−2.232	0.087
S8-t358	−1.623	−0.612	−1.624	0.059	−2.376	−1.218	−2.383	0.088

Table 13
a–c Testing performance indicators for classifiers based on vertical displacement profile at time instant **a** t34, t134 and t234, **b** t284, **c** t358.

Structural health classes	STFT-CNN			WT-CNN			ML		
	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)
Safe-t34	100	100	100	100	100	100	100	100	100
S8-t34	100	100	100	100	100	100	100	100	100
Accuracy (%)		100			100			100	

Structural health classes	STFT-CNN			WT-CNN			ML		
	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)
Safe-t284	100	100	100	100	80.0	88.9	100	100	100
S8-t284	100	100	100	83.3	100	90.9	100	100	100
Accuracy (%)		100			90			100	

Structural health classes	STFT-CNN			WT-CNN			ML		
	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)
Safe-t358	100	100	100	62.5	100	76.9	87.5	70.0	77.8
S8-t358	100	100	100	100	40.0	57.1	75.0	90.0	81.8
Accuracy (%)		100			70			80	

Table 14
Comparative predictive accuracy (%) of all classifiers for the six PdM case studies.

	Accuracy					
	Gap Value			Vertical Displacement		
Predictive Case	STFT-CNN	WT-CNN	ML	STFT-CNN	WT-CNN	ML
3.1	100	93.3	93.3	100	90.0	95.0
3.2	100	90.0	86.7	100	93.3	86.7
3.3	100	95.8	95.8	98.3	97.5	97.5
3.4	100	82.5	97.5	100	88.3	93.3

	Vertical Displacement measured at different sleepers			
	STFT-CNN	WT-CNN	ML	S*
Accuracy Case 3.5	100	85	95	S8
	100	70	100	S11
	100	75	100	S14

	Profiles of vertical displacement for different time instants			
	STFT-CNN	WT-CNN	ML	t*
Accuracy Case 3.6	100	100	100	t34
	100	100	100	t134
	100	100	100	t234
	100	90	100	t284
	100	70	80	t358

misclassification.

In the bolt preload scenarios (Cases 3.1 and 3.2), degradation is detected with high reliability using both gap value and vertical displacement indicators. Importantly, several classifiers exhibit a conservative bias, whereby safe or warning states are occasionally classified as more severe conditions. In predictive maintenance applications, this behaviour is preferable to underestimation of critical damage, as early inspection triggers enhance safety and reduce the likelihood of catastrophic failure. The analysis therefore extends beyond statistical validation and explicitly links error directionality to operational risk management.

The STFT-CNN demonstrates the most consistent performance across scenarios, suggesting that fixed-resolution time–frequency representations effectively capture degradation signatures associated with stiffness reduction. The WT-CNN, while powerful, shows slightly higher sensitivity to subtle waveform similarities in certain configurations, highlighting that representation choice influences automation robustness. This reinforces the importance of selecting feature transformations that are not only accurate but also stable under varying mechanical conditions.

4.1.1. *Interpretability of CNN-based diagnostics*

While deep CNNs are often perceived as black-box models, the present STFT/WT-CNN framework demonstrates a clear correspondence between classification outcomes and physically interpretable signal features. Time–frequency representations of vertical displacement and gap values reveal that the network autonomously captures mechanically meaningful patterns, such as amplitude amplification at wheel–joint interaction points (P1 and P3) and monotonic variations associated with bolt preload reduction or sleeper stiffness loss. Misclassifications, when they occur, are largely confined to adjacent degradation states and can be directly explained by the subtle overlap of corresponding mechanical signatures. This linkage between learned features and structural behaviour provides operational transparency and supports automated predictive maintenance decisions grounded in physical principles, reinforcing the practical applicability of the DT–AI framework in railway infrastructure monitoring.

4.2. *Coupled degradation mechanisms and diagnostic ambiguity*

Case 3.3 reveals that simultaneous bolt loosening and sleeper stiffness degradation may produce partially overlapping signatures in commonly monitored indicators such as gap value and vertical displacement. This finding underscores the necessity of integrated Digital Twin modelling. When degradation mechanisms are analysed in isolation, automated systems risk misattributing structural changes to incorrect sources.

The framework's ability to classify combined defect scenarios confirms that consistent IRJ–sleeper coupling within the Digital Twin enhances diagnostic reliability. At the same time, the observed confusion between certain bolt-loosening configurations and sleeper degradation states reflects an intrinsic physical limitation. When mechanical effects are genuinely similar, no classifier can reliably distinguish them without additional sensing modalities or contextual information. Recognising and explicitly communicating these limits is fundamental to responsible

automation, preventing overconfidence in AI-based decision support.

4.3. Detectability limits and physics-informed sensor strategies

Cases 3.4–3.6 provide insight into the spatial limits of detectability for sleeper stiffness degradation. Stiffness loss in sleepers adjacent to the bonded IRJ generates strong and consistent signatures, whereas degradation located further away becomes progressively more difficult to identify. This attenuation of detectability is governed by structural mechanics rather than algorithmic deficiencies.

From an Automation in Construction perspective, this observation has direct implications for monitoring design. It demonstrates that automation reliability is spatially dependent and that sensor placement strategies must be informed by physics-based understanding of load-transfer mechanisms. The sensor-placement analyses (Cases 3.5 and 3.6) quantify the reduction in classification performance with increasing distance from the damaged sleeper, providing guidance for balancing instrumentation cost and monitoring coverage. The framework therefore supports responsible automation, where decision-makers are informed not only about detected anomalies, but also about zones of low sensitivity where manual inspection or complementary monitoring techniques remain necessary.

The analysis of spatial profiles (Case 3.6) at selected time instants further reveals that diagnostic accuracy peaks when the wheelset passes directly over the degraded sleeper and declines as temporal offset increases. This finding supports the definition of optimized acquisition windows around critical structural locations, reducing data volume, energy consumption, and computational burden. Such optimization is essential for scalable and sustainable PdM deployment in large railway networks. This monitoring paradigm is consistent with emerging image-based sensing approaches [56] and lidar-like technologies capable of capturing deformation fields over extended track sections [62,63].

4.4. Role of digital twins in scalable predictive maintenance

The use of DT-generated data is a deliberate methodological choice aligned with early-stage development of automated monitoring systems. In operational railway networks, fault data are typically scarce, unsafe to reproduce experimentally, or strongly imbalanced toward healthy conditions. High-fidelity Digital Twins provide a controlled environment in which degradation scenarios can be systematically generated, labelled, and used to train AI models.

Importantly, the framework is not intended to replace field measurements, but to bootstrap automation in data-scarce contexts. By first establishing intrinsic separability of mechanically interpretable degradation states, the approach creates a robust baseline for subsequent integration of laboratory and field data. As monitoring datasets become available, the architecture supports hybrid retraining, transfer learning, and progressive calibration. This staged DT–AI synergy enables a gradual transition from simulation-driven validation to data-enriched operational deployment, supporting long-term infrastructure management. This is consistent with DT-based adaptive monitoring frameworks proposed in recent literature [64].

Within this controlled validation setting, the high predictive performance arises from the combined effect of physically structured DT-generated data and the learning capability of the adopted AI models. However, their relative contributions cannot be quantitatively disentangled at this stage. The synthetic datasets are inherently structured, mechanically consistent, and noise-free, which promotes strong class separability; at the same time, effective exploitation of such structured information requires models with sufficient representational capacity. Therefore, while both components are essential, it is not currently possible to quantify how much of the observed performance is attributable to the data structure versus the model architecture. This constitutes a limitation of the study.

Nevertheless, prior experimental evidence reported in [16], where comparatively simple models trained on DT-generated data achieved consistent performance on real measurements, provides initial empirical support for the generalization capability of the modelling approach. The present limitation is therefore not related to model capability per se, but to the absence of systematic experimental benchmarking under realistic operating conditions. Future work will address this aspect through integration of laboratory and field data, enabling explicit assessment of performance degradation under noise, variability, and operational uncertainty, and thus a clearer quantification of the respective roles of data fidelity and model complexity.

4.5. Computational feasibility and deployment considerations

Beyond methodological robustness, practical feasibility is essential for real-world implementation. The selected DCNN architectures were intentionally designed to remain computationally lightweight, favouring scalability and deployment compatibility over marginal accuracy gains obtainable from deeper networks. Processing requirements for STFT and WT transforms, as well as CNN inference, are compatible with contemporary edge- and cloud-enabled monitoring infrastructures.

Gap values and vertical displacements, selected for their strong diagnostic relevance, can be measured using established railway sensors already implemented in practice [17,26]. Within an IoT-enabled monitoring framework, sensors can operate intermittently during train passages, with local preprocessing and transmitting data periodically to edge workstations or remote laboratories, thus minimizing energy consumption while leveraging available computational resources. The structured nature of DT-generated training data further reduces required dataset sizes, lowering both computational and memory demands.

Although full-scale real-time optimization and hardware integration remain subjects of ongoing development, the proposed architecture is explicitly conceived to be compatible with realistic monitoring ecosystems, ensuring technical feasibility of transition from DT-based validation to operational automation.

4.6. Implications for construction automation and lifecycle management

Across all scenarios, the STFT-CNN emerges as the most consistently robust diagnostic approach, while the traditional ML classifier remains an effective and computationally efficient alternative when informative features are known a priori. A central distinction concerns feature engineering: ML methods depend on expert-selected indicators, whereas DL architectures autonomously extract discriminative features from structured time–frequency representations. This autonomy reduces dependence on manual interpretation and supports scalability across diverse degradation scenarios.

By integrating physics-based modelling, AI-driven classification, and automation-oriented interpretation, the proposed framework advances the broader objective of digitally enabled lifecycle management of railway infrastructure. Rather than acting as a black-box fault detector, it explicitly links classification outcomes to mechanical degradation mechanisms, supporting transparent and defensible maintenance decisions.

For infrastructure managers, this translates into:

- automated prioritization of inspections,
- reduced reliance on heuristic thresholds,
- informed sensor-placement strategies,
- and a clearer understanding of when automation is reliable and when human intervention remains essential.

In this sense, the study represents an important step toward responsible, interpretable, and scalable automation in railway predictive maintenance, in line with current research directions in DT-enabled infrastructure management.

4.7. Limitations and structured development pathway

Several limitations should be acknowledged to contextualize the findings and delineate the future development trajectory. First, validation is conducted on Digital Twin–generated synthetic datasets. This choice should be interpreted as a staged validation strategy, in which controlled simulations are used to establish physically grounded feature separability prior to the introduction of real-world uncertainties. While this controlled approach ensures mechanically consistent degradation modelling, it does not explicitly incorporate environmental variability, measurement noise, or operational disturbances typical of in-service conditions. Although previous results [16] provide initial evidence of transferability to experimental measurements, future work will focus on hybrid virtual–physical strategies integrating laboratory and field data through transfer learning and domain adaptation.

Second, the structural modelling adopts a quasi-static finite element formulation with constant wheel speed. This approach captures dominant load-transfer mechanisms governing long-term IRJ behaviour while maintaining computational efficiency, but does not explicitly model transient dynamic effects or speed variability. Extending the framework to dynamic analyses represents a logical progression that may enrich the diagnostic feature space while preserving the physics-informed foundation.

Finally, while computational compatibility has been demonstrated conceptually, full-scale hardware optimization, energy harvesting integration, and real-time field validation remain future research efforts. The present study should therefore be viewed as a structured intermediate phase: it establishes the DT–AI synergy, defines mechanically interpretable health classes, quantifies detectability limits, and demonstrates deployment-oriented feasibility. These elements provide a reproducible and mechanistically grounded foundation upon which progressively data-enriched, experimentally validated, and operationally robust monitoring systems can be built.

5. Conclusions

This paper presented the development and validation of a physics-informed Digital Twin–Artificial Intelligence (DT–AI) framework for automated Structural Health Monitoring (SHM) and Predictive Maintenance (PdM) of railway infrastructure, with specific application to bonded Insulated Rail Joints (IRJs) and adjacent sleepers. Beyond demonstrating high classification accuracy, the principal contribution lies in establishing a mechanically grounded, automation-oriented decision pipeline in which degradation mechanisms are systematically modelled, physically interpreted, and diagnosed through data-driven analytics where high predictive performance emerges from the synergy between physics-informed Digital Twin-generated data and AI-based models rather than from either component in isolation.

5.1. Scientific and methodological contributions

The research advances the state of knowledge along three complementary directions.

First, it proposes a unified DT–AI integration strategy in which a high-fidelity finite element Digital Twin reproduces the coupled mechanical interaction between IRJs and adjacent sleepers under controlled bolt-preload loss and support-stiffness degradation scenarios. Unlike approaches that treat simulation and AI as sequential or loosely connected components, the present framework embeds validated mechanical behaviour directly within the data generation process. Supervised learning is therefore performed on physically interpretable degradation states rather than abstract signal clusters, strengthening transparency and reproducibility.

Second, the study demonstrates that time–frequency deep learning architectures, particularly the STFT-CNN, effectively capture non-stationary mechanical signatures associated with progressive stiffness reduction. By transforming vertical displacement and gap value signals into spectrogram-based representations, the approach eliminates manual feature engineering while preserving mechanical interpretability. Across six progressively complex case studies, the STFT-CNN consistently exhibits stable and conservative classification behaviour, reinforcing its suitability for safety-critical automation contexts.

Third, the work explicitly quantifies detectability limits governed by structural mechanics rather than algorithmic performance. The results confirm that bolt preload degradation and adjacent sleeper stiffness loss generate robust and distinguishable signatures, whereas remote sleeper degradation exhibits progressively reduced sensitivity. By linking classification performance to physical load-transfer mechanisms, the study shifts the focus from accuracy benchmarking toward automation-informed diagnostic reliability, an essential requirement for deployable monitoring systems.

5.2. Relevance to automation in construction and infrastructure management

The proposed framework contributes to digital infrastructure research by establishing a closed-loop DT-enabled PdM workflow in which physics-based modelling directly supports automated decision-making. The systematic use of mechanically consistent synthetic datasets addresses the scarcity and imbalance of real degradation data, while the quasi-static modelling strategy ensures computational efficiency compatible with edge- and cloud-based monitoring architectures.

Importantly, the framework does not operate as a black-box fault detector. Classification outcomes are explicitly traceable to underlying degradation mechanisms, supporting transparent inspection prioritization, physics-informed sensor-placement strategies, and responsible automation within lifecycle asset management.

5.3. Future research directions

Building on the structured validation phase established herein, future developments will focus on four interrelated directions:

1. *Dataset expansion and realism enhancement*, through integration of laboratory and field measurements and the adoption of transfer learning and domain adaptation techniques to bridge synthetic and operational data distributions.
2. *Model generalization and dynamic extension*, including incorporation of variable-speed and transient dynamic simulations to enrich the feature space and improve early-stage degradation sensitivity.
3. *Scalable sensor-network optimization*, aimed at defining minimal yet reliable sensing configurations that balance monitoring coverage, energy consumption, and computational cost.
4. *Integration within digital construction and asset-management ecosystems*, coupling the DT–AI diagnostic module with higher-level digital twin platforms and maintenance planning tools to enable real-time lifecycle decision support.

These directions define a structured pathway from controlled virtual validation toward adaptive, field-ready predictive maintenance systems.

5.4. Ongoing research infrastructure

In parallel with the research activities presented in this study, a dedicated experimental infrastructure has been developed to support the subsequent validation phases of the proposed DT–AI framework. Specifically, the Railway Superstructure Test Frame (RSTF) prototype [65] was constructed within the activities of the Italian project MOST — Centro Nazionale per la Mobilità Sostenibile [4] at the University of Parma under the the National Recovery and Resilience Plan (NRRP) and the Next Generation EU (NGEU) programme [5].

The RSTF constitutes a complex full-scale laboratory apparatus

designed to reproduce controlled dynamic loading conditions representative of railway traffic while accommodating real track components, including rails, ballast, and sub-ballast layers. This experimental platform will enable systematic testing of bonded IRJs and adjacent sleepers under controlled degradation scenarios, thereby generating controlled experimental datasets suitable for direct comparison with the synthetic data produced by the Digital Twin.

Within the broader research programme, the RSTF will therefore provide the experimental foundation for the next development stages of the proposed framework, supporting calibration, cross-validation, and progressive integration of physics-informed AI diagnostics with laboratory-scale infrastructure monitoring.

Together with the ongoing development of dedicated experimental infrastructures, these results establish a structured pathway toward automated predictive maintenance systems for next-generation railway infrastructure monitoring.

CRediT authorship contribution statement

G. Bianchi: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **F. Freddi:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **F. Giuliani:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.autcon.2026.107102.

Data availability

The data that has been used is confidential.

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